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Resolving synaptic events using subsynaptically targeted GCaMP8 variants

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eLife Assessment

In this **important** study, the authors engineered and characterised novel genetically encoded calcium indicators (GECIs) and an analytical tool (CaFire) capable of reporting and quantifying various sub-synaptic events, including miniature synaptic events, with a speed and sensitivity approaching that of intracellular electrophysiological recordings. They present **compelling** data validating this toolset, which will be of interest to neurobiologists studying synaptic calcium dynamics in various model systems.

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Abstract

While genetically encoded Ca^{2+} indicators are valuable for visualizing neural activity, their speed and sensitivity have had limited performance when compared to chemical dyes and electrophysiology, particularly at synaptic compartments. We addressed these limitations by engineering a suite of next-generation GCaMP8-based indicators, targeted to presynaptic boutons, active zones, and postsynaptic compartments at the *Drosophila* neuromuscular junction. We first validated these sensors to be superior to previous versions and synthetic dyes. Next, we developed a Python-based analysis program, *CaFire*, which enables the automated quantification of evoked and spontaneous Ca^{2+} signals. Using *CaFire*, we show a ratiometric presynaptic GCaMP8m sensor accurately captures physiologically-relevant presynaptic Ca^{2+} changes with superior sensitivity and similar kinetics compared to chemical dyes. Moreover, we test the ability of an active zone-targeted, ratiometric GCaMP8m sensor to report differences in Ca^{2+} between release sites. Finally, a newly engineered postsynaptic GCaMP8m, positioned near glutamate receptors, detects quantal events with temporal and signal resolution comparable to electrophysiological recordings. These next generation indicators and analytical methods demonstrate that GCaMP8 sensors, targeted to synaptic compartments, can now achieve the speed and sensitivity necessary to resolve Ca^{2+} dynamics at levels previously only attainable with chemical dyes or electrophysiology.

Introduction

Evolution's ingenuity is evident in its exploitation of the calcium ion (Ca^{2+}) as a potent secondary messenger, essential for controlling a diverse array of physiological events in every cell. The basis for Ca^{2+} 's extraordinary signaling capacity lies in a formidable concentration gradient – maintained at nanomolar levels inside cells versus millimolar outside – that exceeds a thousand-fold. This carefully maintained difference allows for controlled Ca^{2+} entry to rapidly amplify cytosolic Ca^{2+} concentrations, in turn activating specific Ca^{2+} -sensitive proteins to initiate vital downstream signaling pathways (Berridge et al., 2003 [↗](#), Clapham, 2007 [↗](#)). In neurons, this Ca^{2+} -driven control system achieves a pinnacle of sophistication. The powerful Ca^{2+} gradient is

harnessed to orchestrate crucial neuronal processes, from shaping synaptic strength at dendritic spines and modulating gene expression in response to activity, to the precise triggering of neurotransmitter release via synaptic vesicle (Dittman et al., 2019 [↗](#), Jackman et al., 2017 [↗](#), McCarthy et al., 2024 [↗](#)). Therefore, the ability to directly observe and interpret these dynamic Ca^{2+} signals in neurons provides a crucial lens, offering profound insights into the complex cellular operations that define neuronal function and flow of information through neural circuits. Given the paramount importance of Ca^{2+} signaling in virtually all aspects of neuronal function, an extensive series of genetically-encoded Ca^{2+} indicators (GECIs) have been engineered with progressively enhanced speed and sensitivity. The archetypal GECI, GCaMP, ingeniously utilized a modified green fluorescence protein (GFP) flanked by the Ca^{2+} binding protein Calmodulin (CaM) and a Ca^{2+} /CaM-binding M13-like peptide (Nakai et al., 2001 [↗](#)). Upon Ca^{2+} binding, a conformational change in the CaM-peptide interaction allosterically modulates GFP fluorescence. While a conceptual breakthrough, early GCaMP iterations suffered from relatively slow kinetics, limited dynamic range, and modest fluorescence changes (Tian et al., 2009 [↗](#)). Consequently, a variety of sophisticated protein engineering strategies were employed to bolster GCaMP performance, leading to significant improvements represented in the GCaMP6 and GCaMP7 series, which offered faster kinetics and higher signal-to-noise ratios (SNRs) (Dana et al., 2019 [↗](#), Chen et al., 2013 [↗](#)). Finally, the subsequent development of the GCaMP8 series marked a substantial leap forward (Zhang et al., 2023 [↗](#)). This enhanced performance was not merely incremental; it resulted from a deep understanding of the molecular mechanisms underlying GCaMP activation, involving meticulous optimization of the CaM/peptide interface for faster Ca^{2+} association and dissociation and other improvements that led to enhanced brightness and efficient conformational coupling (Zhang et al., 2024 [↗](#)). These comprehensive engineering efforts yielded GCaMP8 variants that exhibit dramatically improved kinetics – both in rise and decay times – and sensitivity relative to previous generations. Indeed, their performance characteristics, including SNR, response speed, photostability, and linearity, became comparable or even superior to some of the best synthetic Ca^{2+} dyes, while retaining the crucial advantage of genetic targetability (Zhang et al., 2024 [↗](#)). These innovations provide an opportunity to resolve Ca^{2+} dynamics at subcellular compartments with unprecedented resolution.

Many studies express GCaMP variants alone in the neuronal cytoplasm, a method that effectively captures the large, relatively slow Ca^{2+} fluxes in soma that correlate with action potential spiking (Akerboom et al., 2012 [↗](#), Peron et al., 2015 [↗](#), Ziv et al., 2013 [↗](#)). However, Ca^{2+} transients within subcellular compartments, such as dendritic spines and presynaptic terminals, are characterized by significantly faster kinetics and occur within smaller, more restricted volumes than those in the general cytoplasm, posing distinct measurement challenges. Importantly, interpreting signals from GCaMPs expressed alone can be confounded by variations in sensor abundance due to uneven expression levels, differing protein stability, or photobleaching, any of which can obscure the accurate determination of true Ca^{2+} changes (McMahon et al., 2018 [↗](#)). Ratiometric indicators are therefore an attractive alternative, as they typically involve co-expression of Ca^{2+} -sensitive and Ca^{2+} -insensitive fluorescent proteins in a stoichiometric manner (Zhang et al., 2021 [↗](#)). By normalizing the Ca^{2+} -dependent signal to the Ca^{2+} -independent one, ratiometric measurements can control for these variations in sensor concentration and provide a more reliable quantification of Ca^{2+} levels and dynamics. Hence, an optimal approach for accurately capturing fast and local subsynaptic Ca^{2+} fluxes would target an advanced, ratiometric GCaMP sensor, engineered for high sensitivity and rapid kinetics, to the particular compartment being investigated.

The *Drosophila* neuromuscular junction (NMJ) is a powerful and genetically tractable model system for dissecting Ca^{2+} signaling dynamics at a glutamatergic synapse. Its utility is magnified by the ability to combine sophisticated genetic manipulation with GECIs, advanced imaging, and established electrophysiological approaches to monitor synaptic events with remarkable spatiotemporal resolution (Bellen et al., 2010 [↗](#), Frank et al., 2013 [↗](#)). Early investigations into Ca^{2+} dynamics at the fly NMJ employed chemical dyes (e.g., OGB-1, Fura-2), which, despite their rapid kinetics, present challenges in controlling loading concentrations at synaptic compartments

(Macleod et al., 2002 [↗](#)). The advent of GECIs offered an opportunity for targeted and consistent expression at subcellular regions. Pioneering engineering of presynaptic GECIs enabled Ca^{2+} visualization at axonal boutons or active zones (Kiragasi et al., 2017 [↗](#), Cohn et al., 2015 [↗](#), Akbergenova et al., 2018 [↗](#)), but were often non-ratiometric for Ca^{2+} sensing – complicating quantification – and employed older GCaMP variants with suboptimal kinetics and SNRs. Postsynaptic sensors evolved from using GCaMP2 to GCaMP6f, offering better sensitivity for detecting quantal events (Newman et al., 2017 [↗](#), Peled et al., 2014 [↗](#)). More recently, GCaMP8f variants were employed (Perry et al., 2022 [↗](#), Li et al., 2021 [↗](#), Han et al., 2023 [↗](#)), demonstrating promising sensitivity. However, other studies suggested that GCaMP8m may offer higher SNRs and similar kinetics, particularly in *Drosophila* (Zhang et al., 2023 [↗](#)). These developments motivated us to engineer and validate optimal GECIs for the fly NMJ, aiming for performance that rivals chemical dyes for presynaptic imaging and electrophysiology for postsynaptic quantal event detection.

We have engineered and characterized a variety of next-generation GECI probes to resolve Ca^{2+} dynamics at pre- and post-synaptic compartments at the *Drosophila* NMJ. While an expected tradeoff between signal strength and kinetics was observed, GCaMP8m consistently delivered the highest responses with only modest reductions in speed relative to GCaMP8f. Notably, presynaptic GCaMP8m sensors captured physiologically important events with speed and sensitivity similar or superior to that of chemical dyes, while postsynaptic GCaMP8m probes achieved quantal event detection with similar resolution to electrophysiological recordings. These features establish targeted ratiometric GCaMP8m sensors and CaFire analysis as complementary tools for resolving synaptic events using purely optical approaches.

Results

Engineering next generation GCaMP sensors targeted to synaptic compartments

To engineer the next generation of synaptically-targeted GCaMP sensors, we sought to improve upon previous sensors employed for Ca^{2+} imaging at the *Drosophila* NMJ. In particular, we focused on three cassettes. First, to target GCaMP to presynaptic boutons, GCaMP6s was fused to the synaptic vesicle protein Synaptotagmin (SYT), generating UAS-SYT::GCaMP6s (Cohn et al., 2015 [↗](#)), schematized in Fig. 1A [↗](#). More recently, this SYT::GCaMP fusion strategy was improved upon by replacing GCaMP6s with the faster and more sensitive GCaMP8f, and rendered ratiometric by fusion to the red-shifted, monomeric fluorescent protein mScarlet (Bindels et al., 2017 [↗](#)), to generate UAS-SYT::mScarlet::GCaMP8f (Scar8f) (Li et al., 2021 [↗](#)) (Fig. 1A [↗](#)). To potentially improve this indicator, we replaced GCaMP8f with GCaMP8m, which is reported to exhibit higher sensitivity with only minor reductions in kinetics (Zhang et al., 2023 [↗](#)), and replaced mScarlet1 with mScarlet3, which encode improvements in brightness and photophysical characteristics (Gadella et al., 2023 [↗](#)), to make the ratiometric GECI Scar8m (Fig. 1A [↗](#)). Expression of each sensor with the motor neuron driver OK319-Gal4 and immunostaining demonstrated the expected trafficking to presynaptic boutons and co-localization with SYT (Fig. 1B [↗](#)).

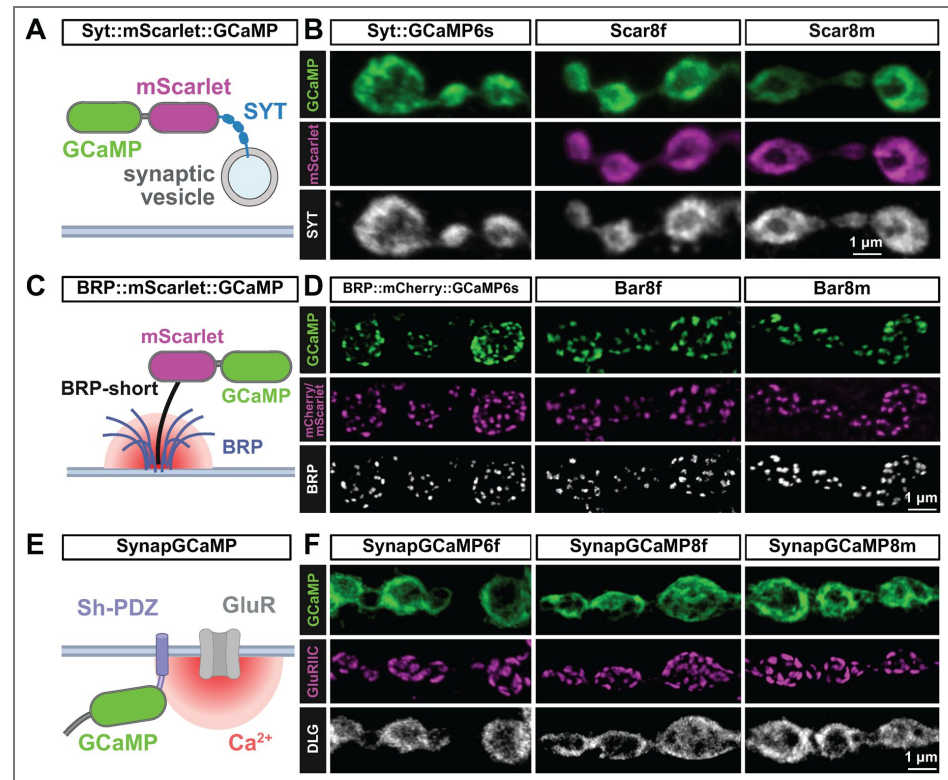


Figure 1. GCaMP indicators targeted to pre- and post-synaptic compartments.

(A) Schematic of the presynaptic ratiometric Syt::mScarlet::GCaMP (Scar8f/Scar8m) Ca^{2+} indicators showing localization to synaptic vesicles via fusion to the Ca^{2+} sensor Synaptotagmin (SYT). (B) Representative images of NMJs expressing the indicated reporter driven in motor neurons with the OK319-GAL4 driver (w^{1118} ; OK319-GAL4/UAS-Scar8f) immunostained with anti-GFP (GCaMP) and anti-SYT. Note that endogenous mScarlet signals were obtained without antibody labeling. (C) Schematic of the BRP::mScarlet::GCaMP8f/8m (Bar8f/Bar8m) ratiometric Ca^{2+} indicator, which targets GCaMP to active zones via fusion to the BRP-short protein (Schmid et al., 2008). (D) Representative images of NMJs expressing the indicated reporter driven in motor neurons (w ;OK6-GAL4/Bar8f and w ;OK6-GAL4/Bar8m) immunostained with anti-GFP (GCaMP) and anti-BRP. Note that native mCherry or mScarlet signals were obtained without antibody labeling. (E) Schematic of the SynapGCaMP indicator, which targets GCaMP to postsynaptic compartments via a Shaker PDZ domain (Newman et al., 2017). (F) Representative images of NMJs expressing the indicated reporter (w ;MHC-CD8-GCaMP6f-Sh/+; w ;MHC-CD8-GCaMP8f-Sh/+; w ;MHC-CD8-GCaMP8m-Sh/+) immunostained with anti-GFP (GCaMP), -GluRIIC (glutamate receptors), and -DLG (postsynaptic density). All confocal images in this figure were deconvolved in Huygens (CMLE) prior to display (see methods).

Next, to target GCaMP to the sites of presynaptic Ca^{2+} influx, active zones (AZs), we improved upon a previous indicator, UAS-BRPshort::mCherry::GCaMP6s (Kiragasi et al., 2017). This strategy used a “short” fragment of the AZ scaffold Bruchpilot (BRP) (Schmid et al., 2008), which traffics to AZs, fused to mCherry and GCaMP6s for ratiometric Ca^{2+} imaging at AZs (Fig. 1C). To improve upon this design, we replaced mCherry with the improved mScarlet, and GCaMP6s with GCaMP8f to make UAS-BRPshort::mScarlet::GCaMP8f (Bar8f) or GCaMP8m and mScarlet3 to make Bar8m (Fig. 1C). Expression of these sensors in motor neurons followed by immunostaining also confirmed the expected trafficking to AZs and co-localization with BRP (Fig. 1D). Finally, we worked to improve upon a postsynaptic Ca^{2+} sensor. A previous cassette called SynapGCaMP6f fused GCaMP6f to a Shaker PDZ motif to target the sensor near postsynaptic NMJ glutamate receptors (GluRs) under the control of a muscle-specific enhancer to enable “quantal imaging” of single synaptic vesicle release events (Newman et al., 2017) (Fig. 1E). A recent improvement replaced GCaMP6f with GCaMP8f (Han et al., 2022), and here we engineered GCaMP8m to make

SynapGCaMP8m (Fig. 1E [↗](#)). Immunostaining of NMJs expressing each sensor confirmed proper trafficking and localization to postsynaptic densities near GluRs (Fig. 1F [↗](#)). In addition, GCaMP is a Ca^{2+} buffer and may therefore impact synaptic development and/or function (McMahon et al., 2018 [↗](#)). Thus, we recorded from NMJs expressing the indicators introduced above and found that expression of the GCaMP sensors did not perturb synaptic transmission, with the exception of Scar8f, which showed a moderate reduction in EPSP amplitudes (Fig. S1). Collectively, these tools established the foundation of our study to go on to benchmark each sensor and determine the optimal indicator, and their speed and sensitivity, to resolve pre- and post-synaptic Ca^{2+} events and dynamics.

Automated analysis of Ca^{2+} imaging data

Current approaches to analyze synaptic Ca^{2+} imaging data either repurpose software designed to analyze electrophysiological data or use custom software developed by groups for their own specific needs, including measuring spike frequency across many neurons (Müller et al., 2012 [↗](#), Giovannucci et al., 2019 [↗](#), Blum et al., 2021 [↗](#), Xing et al., 2018 [↗](#)). We therefore developed a new Python-based software platform capable of automated detection and quantification of synaptic Ca^{2+} events. This program, which we named “CaFire”, analyzes imaging data followed by optional processing such as deconvolution (Fig. 2A [↗](#)). The Ca^{2+} imaging data are then imported into ImageJ to select Regions of Interest (ROIs), which are finally exported to a graphical program plotting event time and intensity data (e.g., Microsoft Excel; see methods for details). The CaFire software reads the Excel file and performs all downstream analysis.

CaFire can analyze both presynaptic evoked and postsynaptic quantal Ca^{2+} events with a user-friendly graphical interface that enables parameter adjustment and event visualization (Fig. 2B [↗](#)). For both quantal and evoked events, peak detection is based on customizable parameters, including threshold amplitude, peak width, and minimum inter-peak distance. These parameters can be adjusted to optimize detection sensitivity and specificity, while misidentified events can be manually corrected. Once events are identified, CaFire then quantifies key parameters, such as peak amplitude ($\Delta F/F$), rise time constants based on an exponential growth formula, and decay time constants using a natural logarithmic decay formula (Fig. 2C [↗](#); see methods). In addition, evoked Ca^{2+} events can be partitioned at different stimulation frequencies (e.g., 1 Hz, 5 Hz) to assess frequency-dependent dynamics. All quantification results are displayed in real-time and can be exported in a tabular format for downstream statistical analysis. This automated workflow significantly reduces analysis time and user variability in Ca^{2+} imaging data analysis when compared to a variety of other platforms we have tried, providing uniform, accurate, and reproducible measurements. CaFire’s ability to analyze both evoked and quantal events makes it a versatile tool for quantifying synaptic Ca^{2+} imaging data.

Benchmarking presynaptic Ca^{2+} sensors

We first sought to directly compare the performance of Scar8f with chemical Ca^{2+} indicator Oregon Green BAPTA-1 (OGB-1) under the same conditions. Images of the type-Ib terminal on muscle #6 (MN6/7-Ib) terminals were captured at 303 frames-per-second on a confocal microscope (Fig. 3A [↗](#)), evoking fluorescence transients at 1 Hz. In all aspects, Scar8f matched or exceeded the performance of OGB-1 (Fig. 3B-E [↗](#)). Specifically, GCaMP8f displayed higher sensitivity compared to OGB-1 and exhibited a faster rate of decay (Fig. 3B,C [↗](#)). The terminals were also challenged with a 21.3 Hz train of stimuli, where individual Ca^{2+} transients were easily discerned in both GCaMP8f and OGB-1 traces (Fig. 3D [↗](#)). However, GCaMP8f fluorescence summated to a far greater degree, representing greater frequency facilitation of the Ca^{2+} transients (Fig. 3D,E [↗](#)). Finally, we compared the evoked Ca^{2+} signals from cytosolic GCaMP8m (RSET-8m, (Reagents et al., 2024 [↗](#))) to Scar8m (Fig. 3F [↗](#)). We observed an almost two-fold increase in the amplitude of the Scar8m response in Scar8m compared to cytosolic GCaMP8m (Fig. 3G,H [↗](#)). Thus, Scar8f/m, targeted to synaptic vesicles, exhibits superior sensitivity and kinetics compared to the chemical dye OGB-1 and cytosolic GCaMP8m indicators.

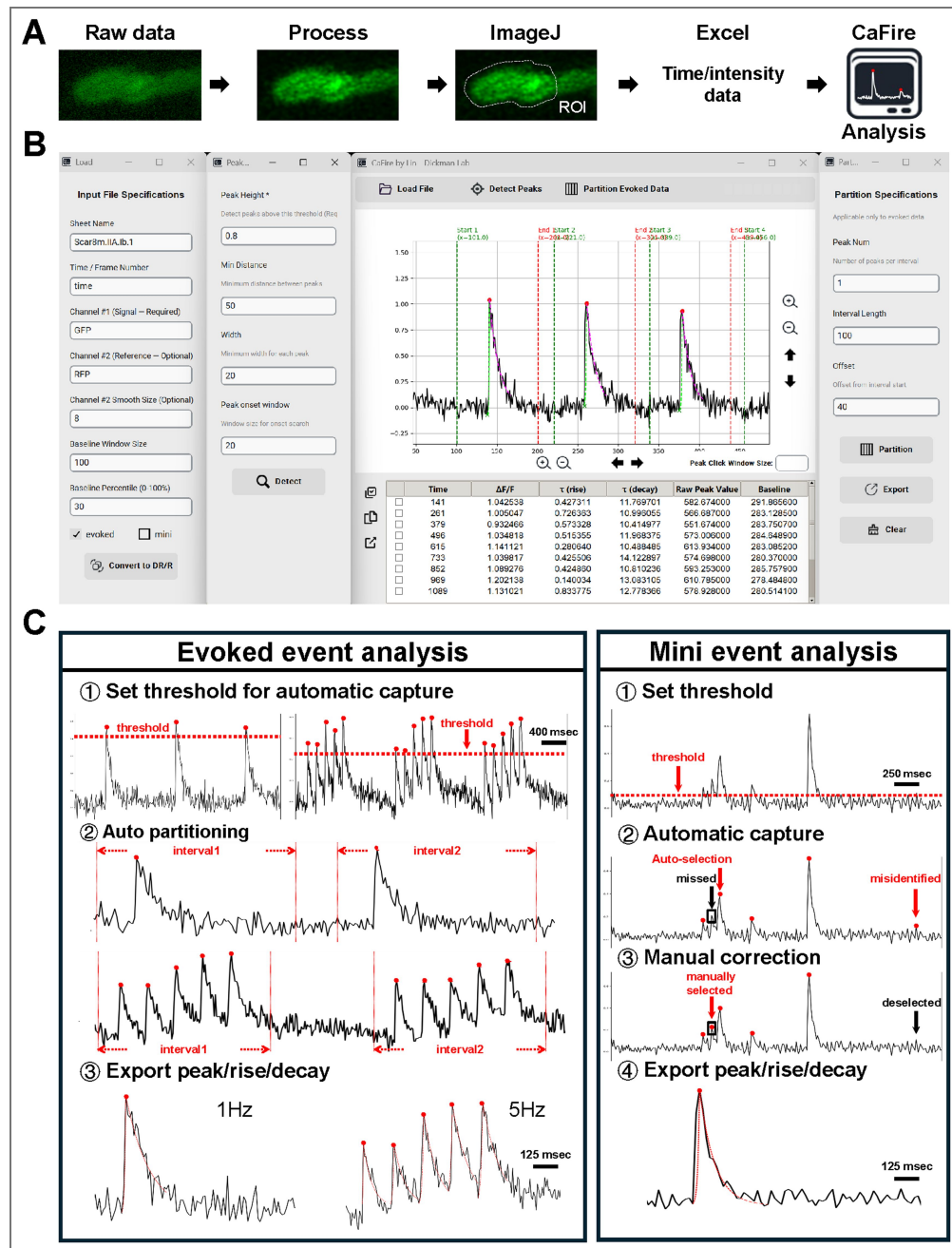


Figure 2. "CaFire" - a Python-based analysis program for quantifying synaptic Ca^{2+} imaging data.

(A) Workflow showing how Ca^{2+} imaging data and downstream analysis is performed. Raw timelapse movies are processed with SVI Huygens software to correct and deconvolve image artifacts. ROIs are then selected in ImageJ Fiji software and intensities are extracted. Data are exported to Excel for manual inspection or directly analyzed using CaFire software. (B) Screenshots of the CaFire user interface. The left panels allow users to specify input file properties and analysis settings, and the second panel from the left shows the peak-detection criteria. The central panel displays fluorescence traces with automatically detected Ca^{2+} events overlaid and the event-based partition results. Detected events and their extracted parameters are listed in the data table below. The right panel shows the partition specifications used to segment evoked responses. (C) Examples of two distinct analysis pipelines implemented in CaFire. *Evoked event analysis*: (1) Thresholds are set for automatic peak detection; (2) Events are automatically partitioned based on stimulus intervals (e.g., 1 Hz and 5 Hz); (3) Parameters such as peak amplitude, rise time constant (τ_{rise}), and decay time constant (τ_{decay}) are calculated using exponential fits. *Mini event analysis*: (1) Users define amplitude thresholds for event detection; (2) CaFire automatically identifies candidate events; (3) Missed or misidentified events can be manually corrected; (4) Event parameters are exported for each validated event.

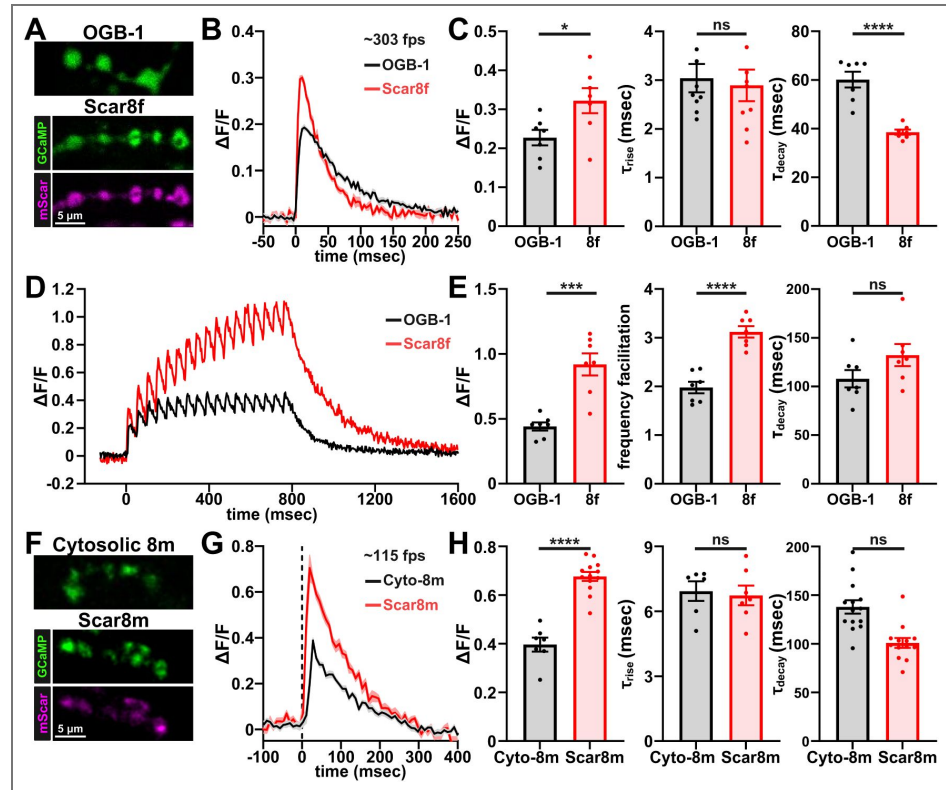


Figure 3. Benchmarking presynaptic GCaMP8 to synthetic dyes and cytosolic sensors.

(A) Confocal images showing presynaptic loading of the synthetic dye OGB-1 and presynaptic expression of Scar8f at MN-Ib boutons. (B) Example single-AP evoked presynaptic Ca^{2+} transients reported by OGB-1 (black) and GCaMP8f (red). Traces represent the average of 30 APs from a single preparation acquired at 303 fps. (C) Quantification of response amplitude ($\Delta F/F$), rise time, and decay times from single-AP Ca^{2+} transients reported by OGB-1 and Scar8f. (D) Example train-evoked Ca^{2+} transients reported by OGB-1 (black) and Scar8f (red) during three trains of 10 APs delivered at 21.3 Hz. (E) Quantification of train-evoked Ca^{2+} signals showing amplitude ($\Delta F/F$) of the final Ca^{2+} transient in the train (17th), measured from baseline; frequency facilitation, calculated as the amplitude of the 17th transient divided by the amplitude of the 1st; and decay time constant (τ_{decay}) measured after the final transient in the train. (F) Confocal NMJs images showing presynaptic expression of the cytosolic sensor RSET-8m (Cytosolic GCaMP8m) and Scar8m. (G) Example single-AP evoked Ca^{2+} transients reported by RSET-8m (black) and Scar8m (red) acquired at ~115 fps. (H) Quantification of amplitude ($\Delta F/F$), rise time (τ_{rise}) and decay time constants (τ_{decay}) from single-AP Ca^{2+} transients reported by RSET-8m (*w;OK319-GAL4/+; UAS-RSET-GCaMP8m/+*) and Scar8m (*w;OK319-GAL4/+; UAS-Syt::mScarlet::GCaMP8m/+*). Shaded traces and bars show mean $\pm$ SEM. Statistical comparisons are unpaired two-tailed t-tests; significance is indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant. Detailed statistics including p-values are provided in Table S1.

Next, we systematically evaluated the performance of the presynaptically targeted Ca^{2+} indicators SYT::GCaMP6s, Scar8f, and Scar8m at MN-Ib terminals. Live confocal imaging confirmed robust expression of all sensors at presynaptic boutons, with colocalized GCaMP and mScarlet fluorescence (Fig. 4A,B). Ca^{2+} transients were elicited by single action potential (AP) electrical stimulation, and fluorescence responses were recorded and analyzed (Fig. 4C,D). Scar8m produced the highest amplitudes of GCaMP/mScarlet ($\Delta R/R$) ratios, with GCaMP8m showing 345.7% higher SNR over GCaMP6s, and 55.7% higher compared to GCaMP8f (Fig. 4D,E). Both Scar8f and Scar8m clearly outperformed GCaMP6s in both sensitivity and speed (Fig. 4D,E), as expected. Importantly, while GCaMP8f exhibited slightly faster decay time constants (66.6 msec vs 99.2

msec), rise time constants were not significantly different between the two sensors (5.3 msec vs 7.0 msec) (Fig. 4E [↗](#)). Thus, Scar8m exhibited an optimal balance between kinetics and signal strength, depending on the objectives of the experimental approach.

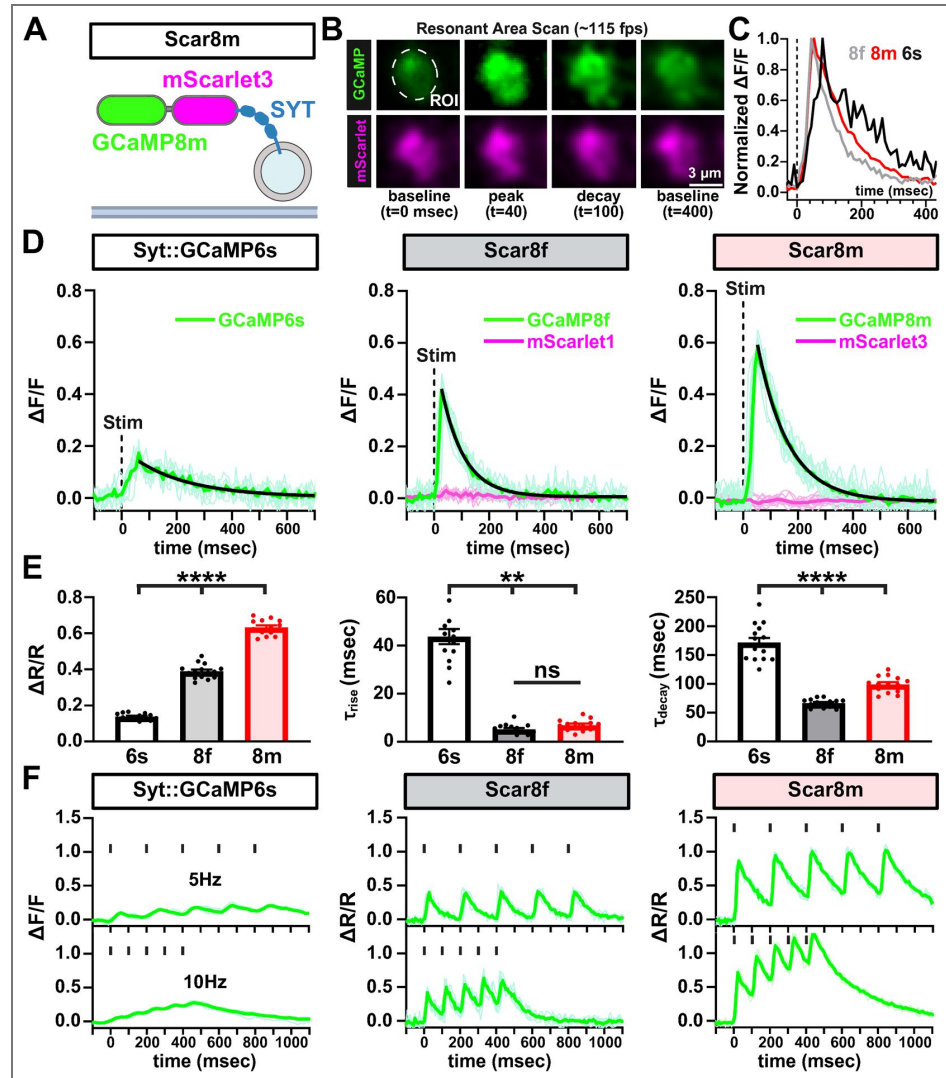


Figure 4. Scar8m is an optimal presynaptic Ca^{2+} indicator.

(A) Schematic of the UAS-Syt::mScarlet3::GCaMP8m (Scar8m) ratiometric Ca^{2+} indicator consisting of mScarlet3 and GCaMP8m targeted to synaptic vesicles via fusion to Synaptotagmin (SYT). (B) Representative images of a MN-Ib bouton expressing Scar8m (*w;OK319-GAL4/+;Scar8m/+*) resonant scanned at ~115 fps. Fluorescence from GCaMP8m (green) and mScarlet3 (magenta) is shown at baseline, peak, decay, and recovery to baseline. (C) Normalized Ca^{2+} signals following single action potential (AP) stimulation. Averaged traces compare the kinetics of GCaMP6s, GCaMP8f, and GCaMP8m. (D) Representative GCaMP and mScarlet signals recorded from Syt::GCaMP6s (GCaMP only), Scar8f, and Scar8m (GCaMP plus mScarlet) in response to single AP stimuli. Thin traces are sequential single-trial sweeps from the same NMJ; the thick colored trace is the mean trace after stimulus alignment. The black curve indicates a single-exponential decay fit used to estimate τ . (E) Quantification of average peak amplitude ($\Delta R/R$, GCaMP/mScarlet ratios), rise time constant (τ_{rise}), and decay time constant (τ_{decay}) from the indicated sensors. Scar8m yields significantly higher peak $\Delta R/R$ signals, similar rise time kinetics, and a modestly slower decay compared to Scar8f. (F) Presynaptic Ca^{2+} responses of the indicated sensors to 5 Hz and 10 Hz stimulation trains. Vertical black ticks above the traces indicate the timing of stimulation pulses. Error bars represent $\pm$ SEM. Significance is indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant; detailed statistics including p-values are presented in Table S1.

To assess sensor performance during repetitive activity, we tested responses to 5 stimuli at 5 Hz and 10 Hz stimulation trains. While responses were muted using SYT::GCaMP6s, Scar8f and Scar8m consistently demonstrated robust responses and separation across repetitive stimulation (Fig. 4F). Together, we conclude that because of the superior sensitivity, ratiometric properties, and fast kinetics, Scar8m is an optimal GECI to quantify stimulated Ca^{2+} responses from presynaptic terminals.

To determine whether the sensitivity of Scar8m is sufficient to quantitatively report biologically relevant differences in evoked presynaptic Ca^{2+} responses, we compared evoked responses between MN-Ib and MN-Is terminals. Previous studies using chemical indicators have shown MN-Is exhibits ~2-3x increased Ca^{2+} responses over MN-Ib (Lu et al., 2016, He et al., 2023). First, we confirmed robust Scar8m expression in both MN-Ib and -Is using the OK319-Gal4 driver (Fig. 5A,B). Next, Scar8m responses to single AP stimulation revealed a >two-fold increase at MN-Is terminals compared to MN-Ib (Fig. 5C,D). Next, we assessed changes in presynaptic Ca^{2+} responses at MN-Ib in wild type and *GluRIIA* mutants, in which a process called presynaptic homeostatic potentiation (PHP) is known to be induced (He et al., 2025, Davis et al., 2015). In PHP, diminished postsynaptic GluR functionality is offset through a homeostatic signaling system which enhances neurotransmitter release (Goel et al., 2021, Delvendahl et al., 2019). PHP induces an increase in presynaptic Ca^{2+} influx to promote additional synaptic vesicle release (Müller et al., 2012, Chien et al., 2025), where chemical indicators such as OGB-1 have shown ~30% increase Ca^{2+} levels after PHP (Müller et al., 2012). Importantly, Scar8m responses in *GluRIIA* mutant MN-Ib boutons showed significantly elevated $\Delta R/R$ amplitudes compared to wild type, with a ~53% enhancement (Fig. 5E,F). Thus, Scar8m can resolve biologically relevant differences in presynaptic Ca^{2+} with high sensitivity, on par or exceeding what chemical indicators have shown.

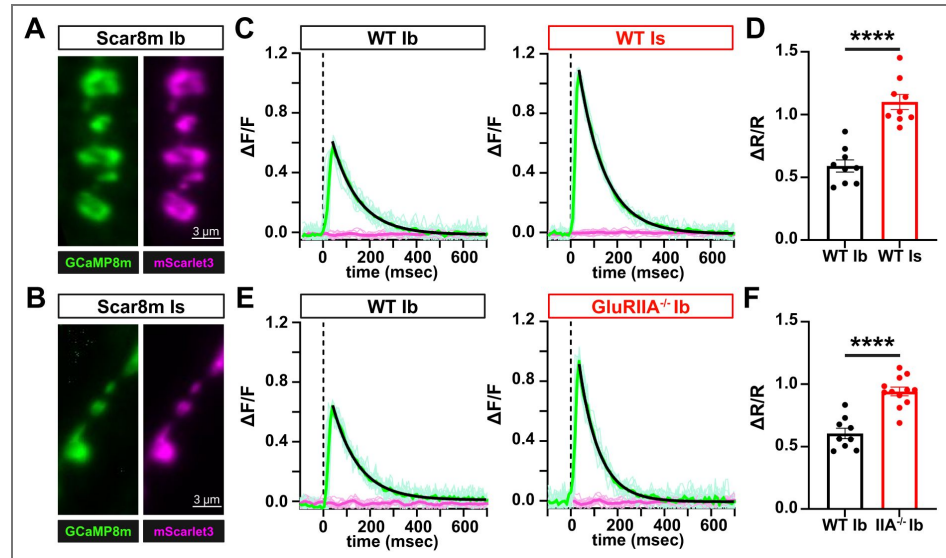


Figure 5. Scar8m captures differences in Ca^{2+} levels between motor neuron subtypes and after plasticity.

(A,B) Representative images of Scar8m expressed at both MN-Ib (A) and MN-Is (B) motor neuron subtypes immunostained with anti-GFP. (C) $\Delta F/F$ traces of GCaMP8m and mScarlet3 responses from single AP stimulation at MN-Ib and MN-Is, with $\sim 2\times$ higher responses observed at MN-Is over -Ib, as expected. Thin traces are sequential single-trial sweeps from the same NMJ; the thick colored trace is the mean trace after stimulus alignment. The black curve indicates a single-exponential decay fit used to estimate τ . (D) Quantification of $\Delta R/R$ responses from the two inputs. (E) $\Delta F/F$ traces of GCaMP8m and mScarlet3 responses from single AP stimulation at MN-Ib in wild type (*w;OK319-GAL4/+;Scar8m/+*) and *GluRIIA* mutants (*w;OK319-GAL4, GluRIIA^{PV3}/GluRIIA^{PV3};Scar8m/+*), which express presynaptic homeostatic plasticity (PHP). Note the enhanced presynaptic Ca^{2+} levels induced after PHP plasticity. (F) Quantification of $\Delta R/R$ responses from the indicated genotypes. Error bars represent $\pm$ SEM. Significance is indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant; detailed statistics including p-values are presented in Table S1.

Evaluating the Bar8m sensor to resolve local active zone Ca^{2+} signals

Differences in Ca^{2+} influx and buffering at individual release sites contribute to functional differences at synapses (Dittman et al., 2019, Jackman et al., 2017, McCarthy et al., 2024). While Scar8m is an excellent sensor to define spatially-averaged presynaptic Ca^{2+} levels at individual boutons, the ability to resolve the heterogeneous Ca^{2+} signals within individual boutons during neurotransmission would be a powerful tool to interrogate presynaptic function. Thus, we engineered the Bar8m sensor to localize a ratiometric GCaMP8m/mScarlet3 cassette to AZs by fusion to the BRP-short motif (Fig. 6A). While some previous studies have suggested a similar approach can resolve AZ-specific differences in Ca^{2+} influx (Akbergenova et al., 2018), from the outset we were cognizant of many challenges to this line of investigation. First, theoretical calculations suggest that Ca^{2+} diffuses and equilibrates at a rate of < 3 msec across an entire bouton, faster than most sensors would respond before receiving Ca^{2+} contributions from other release sites (Maravall et al., 2000, Sinha et al., 1997, Helmchen et al., 1997). Second, it is difficult to resolve individual AZs from live imaging of boutons, as many are closely spaced when flattened 2D images are taken. Finally, at the *Drosophila* NMJ, individual AZs labeled by BRP are very small, with diameters averaging ~ 300 nm (Ramesh et al., 2021). To spatially resolve these discrete AZs during Ca^{2+} imaging, high spatial resolution is required, which often necessitates reducing the scanning rate. Furthermore, when employing dual-channel imaging to

simultaneously capture signals from GCaMP and mScarlet, sequential scanning is typically implemented to minimize spectral crosstalk between channels. These technical constraints collectively limit the achievable frame rate for AZ-level Ca^{2+} imaging to ~ 120 frames per second (fps) using resonant area scanning (Chen et al., 2024 [DOI](#)), although faster frame rates can be achieved with line scanning or CMOS cameras.

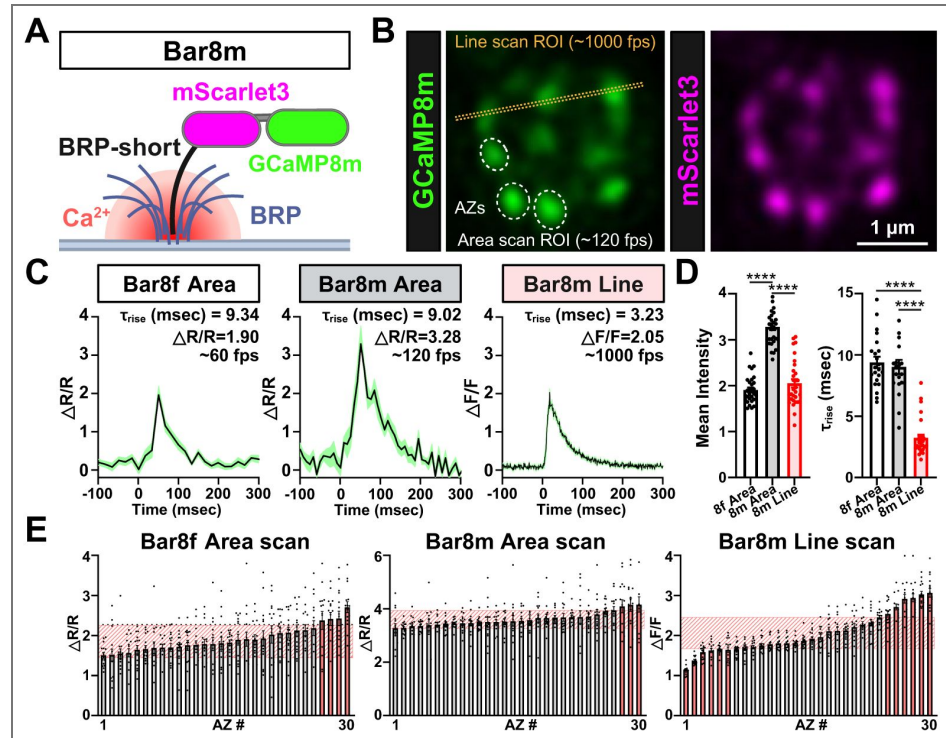


Figure 6. Evaluating the ability of Bar8f and Bar8m to capture active zone-specific Ca^{2+} changes.

(A) Schematic of the BRP::mScarlet3::GCaMP8m (Bar8m) ratiometric Ca^{2+} indicator, consisting of GCaMP8m and mScarlet3 fused to the BRP-short domain, which traffics to individual AZs. (B) Live confocal image of a single MN-Ib bouton expressing Bar8m. Dashed circles indicate active zones used for area-scan measurements, and the yellow dotted line indicates the line-scan ROI used for high-speed imaging. (C) Example Ca^{2+} transients from individual AZs recorded using Bar8f resonant area scans (~60 fps; left), Bar8m resonant area scans (~120 fps; middle), and Bar8m line scans (~1000 fps; right). Black traces show the mean ratiometric response ($\Delta R/R$) for Bar8f and Bar8m area scans and the mean $\Delta F/F$ response for Bar8m line scans from a representative NMJ; green shading denotes $\pm$ SEM across repeated trials. (D) Quantification of mean evoked response amplitude and τ_{rise} for the three imaging conditions shown in (C). (E) Bar graph showing peak $\Delta R/R$ (or $\Delta F/F$) values from individual AZs collected across multiple boutons from different NMJs for Bar8f area scans, Bar8m area scans, and Bar8m line scans. Each bar represents a single active zone (sorted by amplitude within each condition). Light red shaded boxes indicate the interquartile range (IQR; Q1–Q3) of individual data points for each active zone, and active zones whose mean responses fall outside this IQR range are highlighted in pink. Significance is indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant; detailed statistics including p-values are presented in Table S1.

Despite these acknowledged challenges and potential confounds, we sought to test the ability of Bar8m to resolve local AZ Ca^{2+} dynamics following single AP stimulation. Individual AZs were identified at MN-Ib boutons using basal mScarlet3 fluorescence, then evoked Ca^{2+} signals were measured using either resonant area or line scanning (Fig. 6B). Peak $\Delta R/R$ values were measured at individual AZs (Fig. 6C), where typical $\Delta R/R$ values at individual AZs ranged from ~1.5–2.5 (Bar8f), 2.0–3.5 (Bar8m) using resonant area or line scanning (Fig. 6D). These results, while taking into account the caveats detailed above, are consistent with the theoretical estimates of Ca^{2+} levels being equilibrated across all AZs before the Ca^{2+} indicator can capture local differences, at least using relatively slow resonant area scanning approaches.

To further assess Ca^{2+} levels at individual AZ captured by the Bar8f/m sensor, we analyzed our imaging data in more detail. We plotted individual AZ $\Delta R/R$ values from lowest to highest using resonant area scanning (Bar8f and Bar8m), or line scanning (Bar8m) (Fig. 6E). While it was clear

that the relatively slow area scanning was likely insufficient to resolve AZ-specific differences in Ca^{2+} levels, with only 3-4/30 AZs showing significant differences, fast line scanning of Bar8m was capable of capturing significant differences in 12/40 AZs in our imaging system. While we interpret the observed relative consistency of $\Delta R/R$ at individual AZs using resonant area scanning to be unable to reliably resolve local Ca^{2+} changes before equilibration within an individual bouton, faster approaches (line scanning or sCMOS cameras) might be able to detect such heterogeneity.

SynapGCaMP8 sensors resolve quantal events at postsynaptic compartments

Finally, we evaluated the performance of SynapGCaMP6f, SynapGCaMP8f, and SynapGCaMP8m sensors to monitor postsynaptic “quantal” events at the *Drosophila* NMJ. These quantal events reflect the rapid ionic influxes that result from the spontaneous release of single synaptic vesicles, which open postsynaptic GluRs and allow passage of Na^+ and Ca^{2+} ions (Grienberger et al., 2012 [↗](#)). Electrophysiological methods are the gold standard to report these mEPSP events, and we were particularly interested in determining whether SynapGCaMP variants are capable of detecting quantal events with similar sensitivity.

We first confirmed that all SynapGCaMP variants expressed well and trafficked to postsynaptic compartments, localizing with the scaffold DLG while encompassing but being distinct from glutamate receptive fields, which is particularly apparent with super resolution STED microscopy (Fig. 7A [↗](#)). Using resonant area scanning at ~115 fps, we imaged and analyzed single quantal Ca^{2+} events for each SynapGCaMP variant. Averaged traces revealed a progressive improvement in response kinetics and sensitivity across the indicator series (Fig. 7B,C [↗](#)). SynapGCaMP6f exhibited modest sensitivity, with $\Delta F/F$ values of 0.27 and relatively slow kinetics ($\tau_{\text{rise}}=21$ msec, $\tau_{\text{decay}}=99$ msec), while SynapGCaMP8f achieved faster responses and higher sensitivity ($\Delta F/F=0.35$; $\tau_{\text{rise}}=14$ msec, $\tau_{\text{decay}}=42$ msec). SynapGCaMP8m exhibited the highest peak amplitude ($\Delta F/F=0.58$), with similarly fast rise times ($\tau_{\text{rise}}=14$ msec) and moderate slowing of the decay ($\tau_{\text{decay}}=67$ msec), striking an optimal balance between speed and sensitivity (Fig. 7C,E [↗](#)). SynapGCaMP quantal signals appeared to qualitatively reflect the same events measured with electrophysiological recordings (Fig. 7D [↗](#)). We conclude that SynapGCaMP8m is an optimal indicator to measure quantal transmission events at the synapse.

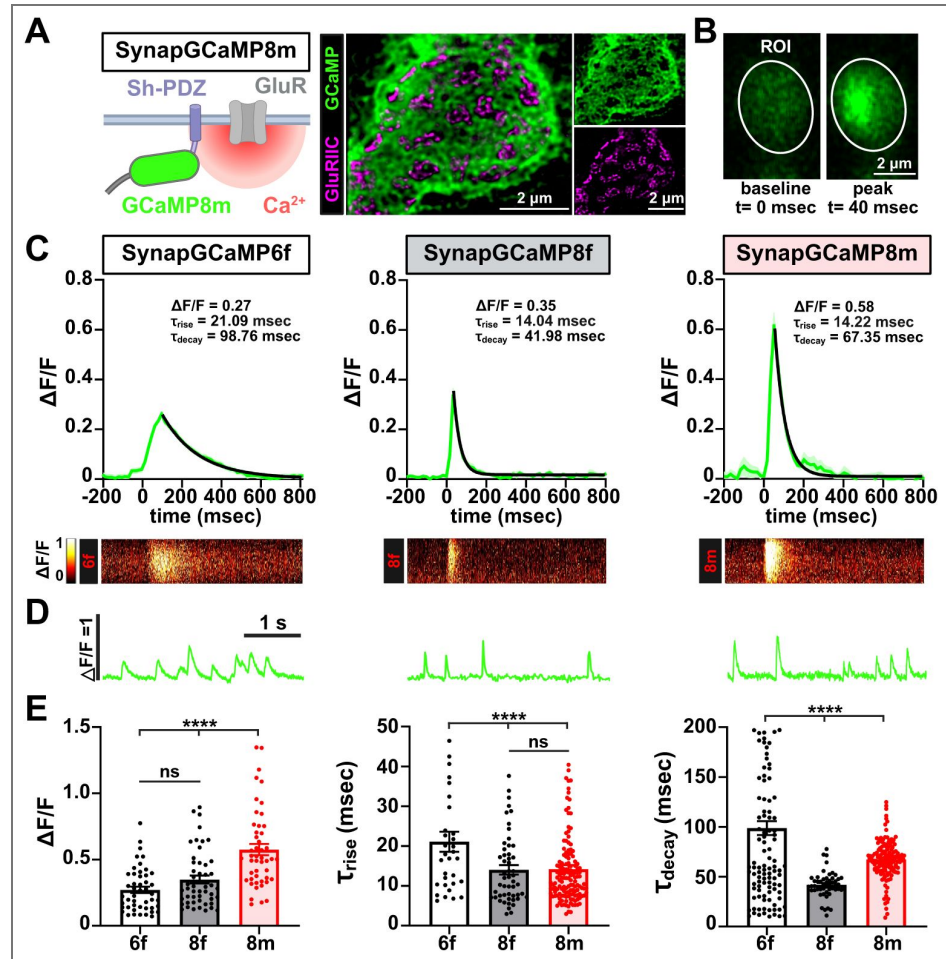


Figure 7. SynapGCaMP8m is an optimal postsynaptic Ca^{2+} indicator.

(A) Schematic of the SynapGCaMP8m reporter, with GCaMP targeted to postsynaptic compartments near glutamate receptors via a Shaker-PDZ motif. Super resolution image using STED microscopy showing the GCaMP8m reporter is localized outside of GluRs. (B) Live confocal images of muscle 6 NMJ boutons expressing SynapGCaMP8m were performed using resonant area scans. The indicated ROI shows representative frames at baseline and peak quantal Ca^{2+} transients acquired at ~ 115 fps. (C) Averaged single miniature Ca^{2+} events recorded from SynapGCaMP6f, -8f, and -8m. The thick green trace shows the mean $\Delta F/F$ waveform and the light green shading indicates $\pm \text{SEM}$; the black curve over the decay phase is a single-exponential fit used to estimate τ_{decay} . SynapGCaMP8m yields the highest peak signal and maintains rapid kinetics. The corresponding heatmaps below were generated from a single vertical line scan extracted from the representative miniature-event ROI, and visualize a spatiotemporal fluorescence dynamics ($\Delta F/F$) along that line over time. (D) Representative Ca^{2+} events reporting individual miniature transmission in the indicated sensors. (E) Quantification of $\Delta F/F$ peak amplitude, and rise and decay time constants (τ_{rise} and τ_{decay}) for each SynapGCaMP variant. All comparisons in bar graphs are statistically significant unless “ns” is shown. Error bars indicate $\pm \text{SEM}$. Significance is indicated as * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant; detailed statistics including p-values are presented in Table S1.

Next, we systematically assessed the ability of the SynapGCaMP variants to resolve quantal events. In particular, we switched to a widefield Ca^{2+} imaging system mounted on an electrophysiology rig so that we could perform simultaneous Ca^{2+} imaging and electrophysiological recordings of mEPSP events. To simplify our analysis, we isolated miniature events from MN-Ib inputs by silencing MN-Is transmission using selective expression of BoNT-C (Han et al., 2022) (see methods). Sharp electrode intracellular recordings and Ca^{2+} imaging from muscle 6 MN-Ib boutons expressing SynapGCaMP6f, -8f, or -8m revealed increasing fidelity of quantal events (Fig.

8A,B [↗](#)). Synchronized GCaMP and mEPSP recordings showed a high degree of temporal correspondence and sensitivity between mEPSPs and Ca^{2+} transients, with SynapGCaMP8m capturing the vast majority of electrophysiologically detected mEPSP events (Fig. 8A,B [↗](#)). Indeed, mEPSP events that failed to register a detectable Ca^{2+} transient were most frequent with SynapGCaMP6f (~40%), rare with SynapGCaMP8f (~10%), and nearly absent with SynapGCaMP8m (~7%) (Fig. 8B [↗](#)). Importantly, when quantal amplitudes from paired mEPSPs and corresponding Ca^{2+} transients were plotted, SynapGCaMP8m exhibited a strong linear correlation with electrophysiological recordings (Pearson's $r = 0.810$, $R^2 = 0.656$; $p < 0.0001$, $n = 216$), demonstrating that the indicator reliably tracks quantal variability (Fig. 8D [↗](#)). SynapGCaMP6f and SynapGCaMP8f also showed significant positive correlations, although weaker than SynapGCaMP8m (6f: $r = 0.458$, $R^2 = 0.210$, $n = 283$; 8f: $r = 0.732$, $R^2 = 0.537$, $n = 298$; all $p < 0.0001$) (Fig. 8C [↗](#)). Thus, SynapGCaMP8m provides a sensitive indicator of quantal events, approaching the sensitivity of electrophysiological recordings.

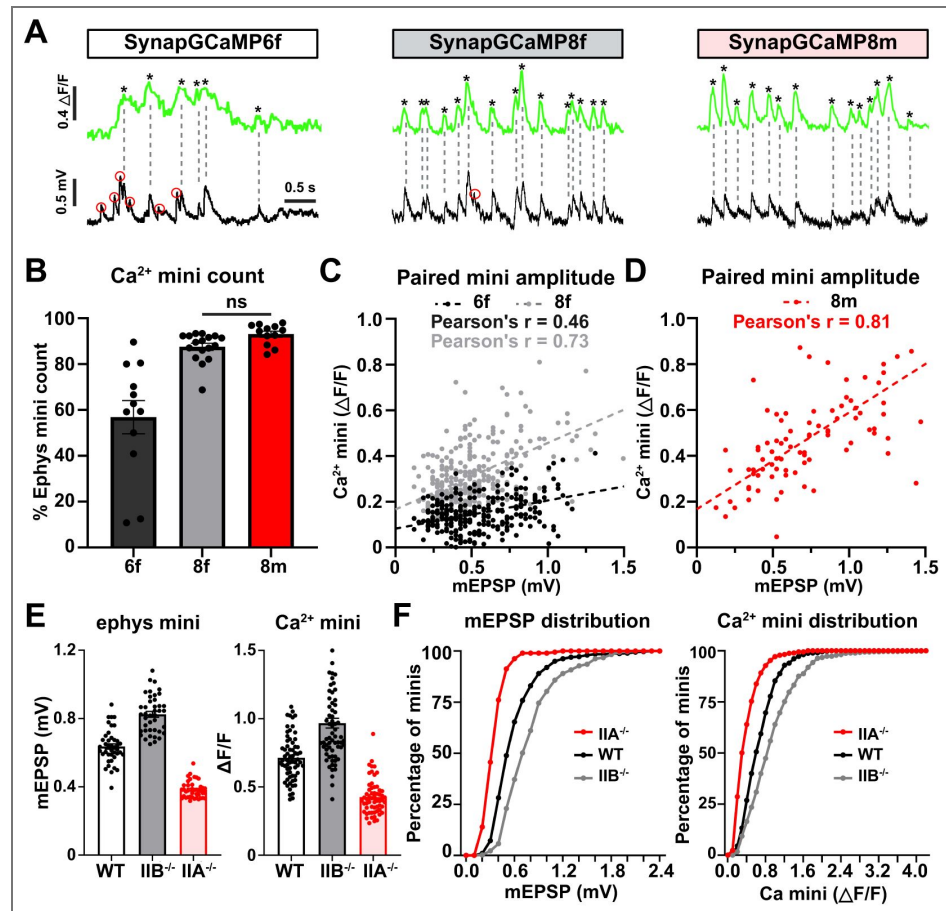


Figure 8. SynapGCaMP8m quantal resolution approaches that of electrophysiology.

(A) Simultaneous recordings of quantal events at MN-Ib boutons using the indicated SynapGCaMP variant (green) and electrophysiology (black) after silencing MN-Is (*w;Is-GAL4/+;UAS-BoNT-C/SynapGCaMP*). Red circles indicate mEPSP events not captured by SynapGCaMP, and asterisks mark Ca^{2+} minis detected optically. SynapGCaMP8m captures quantal events with high sensitivity, comparable to electrophysiology. (B) Quantification of the proportion of quantal events captured by the indicated SynapGCaMP variant as a proportion of the total mEPSP events recorded by electrophysiology. SynapGCaMP6f detects only about half of electrophysiological events, while both SynapGCaMP8f and SynapGCaMP8m capture nearly all mEPSPs. (C) Scatter plot of paired miniature event amplitudes recorded simultaneously by SynapGCaMP6f and -8f and electrophysiology. Each point represents a single matched event; Pearson's correlation coefficients ($r=0.46$ for 6f and $r=0.73$ for 8f) are indicated. (D) Scatter plot of paired miniature event amplitudes recorded simultaneously by SynapGCaMP8m and electrophysiology. Each point represents a single matched event; a strong linear relationship is observed (Pearson's $r=0.81$), indicating that optical signals scale proportionally with quantal amplitude. (E) Bar plots showing average mEPSP amplitudes (left) and $\Delta F/F$ amplitudes of quantal Ca^{2+} events (right) in the indicated genotypes (same conditions as in (A) but including *GluRIIA* or *GluRIIB* mutant alleles). SynapGCaMP8m accurately resolves quantal size differences with similar resolution as the electrophysiological data, with quantal amplitudes in both datasets exhibiting the expected relationship ($\text{GluRIIA}^{-/-} < \text{WT} < \text{GluRIIB}^{-/-}$). Bars show mean $\pm$ SEM, dots represent individual boutons. (F) Cumulative frequency distributions of mEPSP amplitudes (left) and Ca^{2+} mini event amplitudes (right); each are significantly different using the Kolmogorov-Smirnov Test. See Table S1 for full statistical details including p-values.

In our final set of experiments, we focused on benchmarking the ability of SynapGCaMP8m to resolve physiologically relevant differences in quantal events compared to electrophysiology. Specifically, we compared electrophysiological recordings and SynapGCaMP8m imaging of quantal

events in three genotypes with known differences in quantal amplitudes: wild type as the baseline control, *GluRIIA* mutants with diminished quantal amplitudes, and *GluRIIB* mutants with enlarged quantal events (Han et al., 2023 [DOI](#)).

Electrophysiological recordings from each of the three genotypes confirmed the expected differences, with baseline mEPSPs from MN-Ib averaging ~0.64 mV, *GluRIIB* mutants ~0.82 mV, and *GluRIIA* mutants ~0.38 mV (Fig. 8E [DOI](#)), as shown in previous studies (Han et al., 2023 [DOI](#), Han et al., 2022 [DOI](#), He et al., 2023 [DOI](#)). Plotting the mEPSP distribution as a cumulative probability histogram also showed the expected distribution of these genotypes (Fig. 8F [DOI](#)). We then performed a similar analysis of quantal events imaged with SynapGCaMP8m. These results demonstrated near identical differences in average quantal amplitude and distribution in quantal events imaged from wild type, *GluRIIA*, and *GluRIIB* mutants (Fig. 8E,F [DOI](#)). In simultaneous SynapGCaMP8m imaging and current-clamp recordings, the vast majority of optical miniature events were accompanied by a corresponding mEPSP, with less than 5% of SynapGCaMP8m transients lacking a clearly detectable mEPSP; these may reflect very small quantal events that fall below the electrophysiological detection threshold (~0.2 mV). These data demonstrate that SynapGCaMP8m can capture physiologically relevant differences in quantal events with similar sensitivity as electrophysiology.

Discussion

This study presents a significant advancement in our capacity to optically interrogate Ca^{2+} dynamics at synaptic compartments of the *Drosophila* neuromuscular junction, approaches that in principle can be extended to other systems. By systematically engineering and rigorously benchmarking a new suite of ratiometric GECIs – presynaptic Scar8m, active-zone targeted Bar8m, and postsynaptic SynapGCaMP8m – and complemented with the development of the CaFire automated analysis platform, we have overcome limitations of previous tools. These indicators provide a powerful toolkit for dissecting the Ca^{2+} signals that mediate and control synaptic transmission and plasticity, moving us closer to all optical interrogation with similar resolution as electrophysiology. These results demonstrate that careful sensor design, incorporating the latest GCaMP8m variant and optimized ratiometric partners like mScarlet3, can yield probes with unprecedented performance in this model system.

Presynaptic Ca^{2+} imaging with Scar8m not only surpasses previous GECIs but also rivals, and in some respects exceeds, the capabilities of traditional synthetic Ca^{2+} indicators. While chemical dyes offer rapid kinetics (Hendel et al., 2008 [DOI](#), Lock et al., 2015 [DOI](#), Macleod, 2012 [DOI](#), Lnenicka et al., 2006 [DOI](#), Justs et al., 2022 [DOI](#), Lu et al., 2016 [DOI](#)), their utility is often hampered by inconsistent loading/concentrations, lack of cell-type specificity, and challenges in ratiometric quantification within the confined space of presynaptic boutons. Indeed, traditional approaches to load dyes into fly motor neurons relies on cutting the nerve while bathed in solution, and waiting for the dye to diffuse the considerable distance to presynaptic terminals, where variation in dye concentration can confound analyses (McMahon et al., 2018 [DOI](#), Macleod, 2012 [DOI](#)). Scar8m, targeted to synaptic vesicle pools at boutons via fusion to Synaptotagmin along with the ratiometric mScarlet3 motif, circumvents these issues, providing robust, genetically targeted, and stoichiometric and quantifiable Ca^{2+} measurements, spatially-averaged and confined to individual boutons that outperform cytosolic GCaMPs (Fig. 3 [DOI](#)). Further, the superior sensitivity and relatively fast kinetics of Scar8f/m enabled the resolution of physiologically meaningful differences in presynaptic Ca^{2+} levels, such as those between MN-Ib and MN-Is terminals and the elevations associated with presynaptic homeostatic plasticity (Li et al., 2021 [DOI](#)). When directly comparing OGB-1 to Scar8f, we found superior sensitivity and kinetics (Fig. 3 [DOI](#)), which, by extrapolation, should also compare favorably to other synthetic dyes like rhod and Fura (Lu et al., 2016 [DOI](#), He et al., 2023 [DOI](#)). Notably, Scar8f achieves a $\Delta F/F$ of approximately 0.32 vs 0.23 for OGB-1, while also showing a faster decay time constant (~40 msec for Scar8f vs. ~60 msec for OGB-1). Furthermore, the rise time performance of Scar8f/m is on par with that of OGB-1, with a single-AP rise time of ~12 msec compared to ~8 msec (Fig. 3 [DOI](#)), with the caveat that scan speed can limit the accuracy

of rise times. This capacity to faithfully report presynaptic Ca^{2+} changes, previously accessible primarily via chemical dyes, underscores Scar8m's power as a tool for investigating presynaptic function with enhanced precision and reliability.

Equally significant is the performance of the postsynaptically targeted SynapGCaMP8m, which we have shown achieves a sensitivity for detecting quantal Ca^{2+} events that approaches the gold standard of electrophysiological recordings. Previous optical methods often struggled to reliably capture these small, stochastic events, particularly in the case of glutamate receptor loss or perturbation (Newman et al., 2017 [DOI](#), Peled et al., 2014 [DOI](#), Han et al., 2022 [DOI](#)), and have never previously been benchmarked simultaneously with electrophysiological recordings to determine their fidelity. We found that SynapGCaMP6f failed to capture over 40% of mEPSP events, while the GCaMP8m version reliably detected over 90% of these events, even in cases of diminished amplitude due to *GluRIIA* loss. Leveraging the high responsivity of GCaMP8m enabled substantially improved peak amplitudes and kinetics. Furthermore, the strong correlation observed between the amplitudes of optical quantal events and their corresponding mEPSPs, along with the ability to accurately resolve known differences in quantal size in *GluRIIA* and *GluRIIB* mutants, serves to validate SynapGCaMP8m as a high-fidelity reporter of postsynaptic activity. Finally, we note the additional benefits of quantal imaging when compared to electrophysiology, where spatial discrimination of quantal events allows, for example, the ability to quantify input-specific events (see Supplemental Video 1). This opens exciting avenues for all-optical interrogation of quantal parameters, synaptic strength, and changes due to plasticity with subcellular spatial resolution.

Our investigation into capturing local Ca^{2+} signals using an active zone-targeted, ratiometric GCaMP8m based indicator, Bar8m, while ambitious, also yielded important insights despite the inherent technical challenges. Bar8m was able to sensitively report Ca^{2+} changes at individual AZs following single AP electrical stimulation, as evidenced by the significant differences observed across AZs, expected by variation in Ca_v2 channel abundance shown in previous studies (Cunningham et al., 2022 [DOI](#), Gratz et al., 2019 [DOI](#), Medeiros et al., 2024 [DOI](#)). However, the Bar8f/m intensity signals showed relative consistency across AZs following single AP stimulation and imaged using resonant area scans. This finding, coupled with theoretical considerations of Ca^{2+} diffusion rates (Sinha et al., 1997 [DOI](#), Helmchen et al., 1997 [DOI](#)), suggests that within the temporal resolution of current GCaMP sensors, and likely synthetic dyes, Ca^{2+} concentrations are likely to equilibrate across AZs within an individual bouton before distinct local concentration differences can be reliably resolved. There are a number of other potential confounds precluding a definitive conclusion about the ability of Bar8f/m to resolve local Ca^{2+} changes, including challenges in resolving individual AZs and imaging speeds. Indeed, more promising results were observed using line scanning, with kHz speeds, of Bar8m, which appeared to capture a higher degree of AZ-specific differences (Fig. 6E [DOI](#)). These observations are critical for interpreting data acquired from AZ-targeted GECIs and guide future efforts to capture localized Ca^{2+} nanodomain dynamics, likely requiring sensors with even faster kinetics and/or alternative imaging modalities.

The suite of next-generation GECIs, particularly Scar8m and SynapGCaMP8m, coupled with the CaFire analysis pipeline, represents a substantial toolkit for studying synaptic Ca^{2+} signaling. At the *Drosophila* NMJ, these tools provide researchers with unprecedented optical access to both presynaptic Ca^{2+} influx driving neurotransmitter release and postsynaptic Ca^{2+} transients reflecting quantal events, with performance characteristics that rival traditional, more invasive methods. The ability to perform ratiometric, genetically-targeted imaging with such high sensitivity and temporal resolution will help to accelerate discoveries into the molecular mechanisms of synaptic transmission, plasticity, and disease etiology. More generally, the strategies used here in the fly system can inspire similar approaches to be employed in other systems. Future studies leveraging these advanced sensors can now tackle complex questions regarding the spatial and temporal dynamics of Ca^{2+} signaling within synaptic compartments, the modulation of these signals during various forms of plasticity, and their dysregulation in models of neurological disorders, further cementing the *Drosophila* NMJ as a premier system for fundamental synaptic research.

Materials and methods

Fly stocks

Drosophila stocks were raised at 25°C using standard molasses food. Unless otherwise specified, the *w¹¹¹⁸* strain was used as the wild-type control as this is the genetic background in which all genotypes were bred. All experiments were performed on *Drosophila* third-instar larvae of both sexes unless otherwise noted. See Table S2 (Key Resources Table) for a full list of all fly stocks and their sources used in this study.

Molecular biology

To generate the transgenic constructs used in this study, we built upon previously established tools. First, SynapGCaMP8m was engineered by modifying the previous SynapGCaMP8f construct (Han et al., 2022 [DOI](#)), which was in turn based on the original SynapGCaMP6f construct (Newman et al., 2017 [DOI](#)). Gibson assembly reactions were used to convert the few relevant amino acid differences between GCaMP8f and GCaMP8m (Zhang et al., 2023 [DOI](#)). Similarly, UAS-Syt::mScarlet3::GCaMP8m (Scar8m) was generated using similar approaches to modify the previously described UAS-Syt::mScarlet::GCaMP8f plasmid (Li et al., 2021 [DOI](#)), changing both the GCaMP8f and mScarlet sequences to GCaMP8m and mScarlet3 (Gadella et al., 2023 [DOI](#)). Finally, UAS-BrpS::mScarlet::GCaMP8f (Bar8f) and UAS-BrpS::mScarlet3::GCaMP8m (Bar8m) were engineered based on the previous UAS-BrpS::mCherry::GCaMP6s transgene (Kiragasi et al., 2017 [DOI](#)), where the GCaMP6s sequence was replaced with GCaMP8f or GCaMP8m, and the mCherry tag was substituted with mScarlet or mScarlet3. SynapGCaMP transgenes were inserted randomly through p-element transposition, while all other transgenes were inserted into the attP40 (II) or VK27 (III) sites by BestGene Inc. (Chino Hills, CA). All constructs were verified by Sanger sequencing prior to injection, and transgenic fly stocks were established and maintained under standard laboratory conditions.

Immunocytochemistry

Third-instar larvae were dissected in ice cold 0 Ca²⁺ HL-3 and immunostained as described (Li et al., 2021 [DOI](#), Han et al., 2023 [DOI](#)). Briefly, larvae were either fixed in 100% ice-cold methanol for 5 mins or PFA for 10 mins followed by washing with PBS containing 0.1% Triton X-100 (PBST). Samples were blocked with 5% Normal Donkey Serum for 1 hour and incubated with primary antibodies overnight at 4°C. Preparations were washed for 10 mins thrice in PBST, incubated with secondary antibodies for 2 hours at room temperature, washed thrice again in PBST, and equilibrated in 70% glycerol. Prior to imaging, samples were transferred in VectaShield (Vector Laboratories, Burlingame, CA) and mounted on glass cover slides. Details of all antibodies and sources are listed in Table S2.

Electrophysiology

All dissections and electrophysiological recordings were performed as described (Kikuma et al., 2019 [DOI](#)) in modified hemolymph-like saline (HL-3) containing (in mM): 70 NaCl, 5 KCl, 10 MgCl₂, 10 NaHCO₃, 115 Sucrose, 5 Trehelose, 5 HEPES, pH=7.2, and CaCl₂ at 0.4 mM unless otherwise specified. Internal guts, brain and the ventral nerve cord were subsequently removed to achieve fully dissected preparations. Recordings were carried out on an Olympus BX61 WI microscope stage equipped with a 40x/0.8 NA water-dipping objective and acquired using an Axoclamp 900A amplifier (Molecular Devices). All recordings were conducted on abdominal muscle 6, segment A3 of third-instar larvae of both sexes. Data were acquired from cells with an initial resting potential between -60 and -80 mV, and input resistances >5 MΩ. Miniature excitatory postsynaptic potentials (mEPSPs) were recorded without any stimulation and low pass filtered at 1 kHz. The mEPSPs for each sample were recorded for 60 secs, analyzed with MiniAnalysis (Synaptosoft), and the average mEPSP amplitude for each NMJ was calculated. Excitatory postsynaptic potentials (EPSPs) were

recorded by delivering 20 electrical stimuli at 0.5 Hz with 0.5 msec duration to motor neurons using an ISO-Flex stimulus isolator (A.M.P.I.) with stimulus intensities set to avoid eliciting multiple EPSPs.

Confocal imaging

Dissections and live Ca^{2+} imaging was performed as described (Chien et al., 2025 [Chien et al., 2025](#) [Chen et al., 2024](#) [Chen et al., 2024](#)) on muscle 6 NMJs. Wandering third-instar larvae were dissected and imaged in 1.8 mM Ca^{2+} HL3 saline. Ca^{2+} imaging was conducted using a Nikon A1R resonant scanning confocal microscope equipped with a 60x/1.0 NA water-immersion objective (refractive index 1.33). GCaMP signals were acquired using the FITC/GFP channel (488-nm laser excitation; emission collected with a 525/50-nm band-pass filter), and mScarlet/mCherry signals were acquired using the TRITC/mCherry channel (561-nm laser excitation; emission collected with a 595/50-nm band-pass filter). ROIs focused on terminal boutons of MN-Ib or -Is motor neurons. For both channels, the confocal pinhole was set to a fixed diameter of 117.5 μm (approximately three Airy units under these conditions), which increases signal collection while maintaining adequate optical sectioning. Images were acquired as 256 $\times$ 64 pixel frames (two 12-bit channels) using bidirectional resonant scanning at a frame rate of $\sim$ 118 frames/s; the scan zoom in NIS-Elements was adjusted so that this field of view encompassed the entire neuromuscular junction and was kept constant across experiments. In ratiometric recordings, the 488-nm (GCaMP) and 561-nm (mScarlet) channels were acquired in a sequential dual-channel mode using the same bidirectional resonant scan settings: for each time point, a frame was first collected in the green channel and then immediately in the red channel, introducing a small, fixed frame-to-frame temporal offset while preserving matched spatial sampling of the two channels. Resonant area scans of Bar8f and Bar8m responses were acquired at $\sim$ 60 or $\sim$ 120 frames/s with scanning area of 256 $\times$ 32–64 pixels, using a zoom factor of 8 $\times$. Because Bar8f was dimmer, the confocal image quality setting was increased from level 1 to 2, which reduced the scanning frame rate. To resolve rapid Ca^{2+} transients at individual AZs, we additionally performed single-channel Galvano line-scan imaging of Bar8m. A brief resonant area image was first acquired to localize BRP-positive puncta, and a one-dimensional scan line was then positioned through the center of each selected AZ. Line-scan images were acquired at $\sim$ 1000 frames/s sampling rate. Presynaptic and active zone Ca^{2+} imaging was conducted for 15 secs per acquisition while delivering electrical stimulation at 1 Hz (1 msec pulse duration) or at 5 Hz and 10 Hz (five pulses per 2-second burst). Action potentials were evoked by extracellular stimulation of the segmental motor nerve using a glass suction electrode. Suction electrodes were pulled from TW120-4 glass capillaries on a Sutter P-97 puller and fire-polished to a final tip opening of $\sim$ 5–10 μm , then filled with HL3 saline. The cut motor nerve innervating the imaged muscle was gently drawn into the electrode, and electric stimulation was delivered through a pulse stimulator coupled to an isolated stimulator (Iso-Flex, controlled by a Master-9 pulse generator)(Chen et al., 2024 [Chen et al., 2024](#)). To prevent muscle contraction during imaging, 7 mM L-Glutamic acid monosodium salt (Sigma-Aldrich, Cat# G5889) was added to the HL3 saline. In contrast, postsynaptic Ca^{2+} imaging was performed for 60 secs without electrical stimulation, as only quantal events were monitored. Data was collected from at least three biological replicates per genotype.

Time-lapse videos were stabilized and bleach-corrected prior to analysis, which visibly reduced frame-to-frame motion and intensity drift. In the presynaptic and active-zone mScarlet channel, a bleaching factor of $\sim$ 1.15 was observed during the 15 sec recording. This bleaching can be corrected using the “Bleaching correction” tool in Huygens SVI. For presynaptic and active-zone GCaMP signals, there was minimal bleaching over these short imaging periods.

Therefore, the bleaching correction step for GCaMP was skipped. Both GCaMP and mScarlet channels were processed using the default settings in the Huygens SVI “Deconvolution Wizard” (with the exception of the bleaching correction option). Deconvolution was performed using the CMLE algorithm with the Huygens default stopping criterion and a maximum of 30 iterations, such that the algorithm either converged earlier or, if convergence was not reached, was terminated at this 30-iteration limit; no other iteration settings were used across the GCaMP series. ROIs were drawn on the processed images using Fiji ImageJ software, and mean fluorescence time

courses were extracted for the GCaMP and mScarlet channels, yielding $F_{\text{GCaMP}}(t)$ and $F_{\text{mScarlet}}(t)$. $F(t)$ s were imported into CaFire with GCaMP assigned to Channel #1 (signal; required) and mScarlet to Channel #2 (baseline/reference; optional). If desired, the mScarlet signal could be smoothed in CaFire using a user-specified moving-average window to reduce high-frequency noise. In CaFire's $\Delta R/R$ mode, the per-frame ratio was computed as $R(t) = F_{\text{GCaMP}}(t)/F_{\text{mScarlet}}(t)$; a baseline ratio R_0 was estimated from the pre-stimulus period, and the final response was reported as $\Delta R/R(t) = [R(t) - R_0]/R_0$, which normalizes GCaMP signals to the co-expressed mScarlet reference and thereby reduces variability arising from differences in sensor expression level or illumination across AZs (see CaFire section in methods for details).

Confocal images of fixed tissue were acquired with a 100x APO 1.45NA oil immersion objective using separate channels with four laser lines (405 nm, 488 nm, 561 nm, and 647 nm) as described (Perry et al., 2017 [DOI](#)). Z-stacks were obtained on the same day using identical gain and laser power settings with z-axis spacing between 0.13 and 0.2 μm and pixel size of 0.06 μm for all samples within an individual experiment, from at least 8 NMJs acquired from at least four different animals. Raw confocal images were deconvolved with SVI Huygens Essential 22.10 using built-in Express settings. All confocal images shown in the figures, including Fig. 1 [DOI](#), were deconvolved in Huygens using this CMLE-based workflow prior to maximum-intensity projection and display. Maximum intensity projections were created for quantitative image analysis using the general analysis toolkit of NIS Elements software.

Benchmarking OGB-1 vs Scar8f

For OGB-1 imaging, wild-type motor neuron terminals were forward filled with Oregon Green 488 BAPTA-1 (OGB-1; 10,000 MW anionic dextran, batch-specific $K_d = 1180$ nM, Molecular Probes) as described (Macleod et al., 2002 [DOI](#)). Larvae were rinsed in distilled water, transferred to Schneider's *Drosophila* medium, and pinned dorsal-side up on Sylgard-coated dishes with Minutien pins. A midline incision was made along the dorsal surface, the body wall was pinned open to generate a fillet preparation, and internal organs were removed to expose the ventral nerve cord and segmental nerves. For forward filling, segmental nerves were severed near the ventral ganglion and the cut ends from hemisegment 4 were drawn into a glass suction pipette; a bolus of 5 mM OGB-1 in distilled water was ejected onto the nerve ends in an equal volume of Schneider's medium, left in place for 30–60 min, then replaced with Schneider's medium in the pipette to serve as the cathode for stimulation. OGB-1 was allowed to transport to NM6/7-Ib terminals for 4–5 h at room temperature, with Schneider's medium exchanged hourly. Immediately prior to imaging, Schneider's medium was replaced with hemolymph-like 6 (HL6) saline containing 2 mM CaCl_2 and 7 mM L-glutamic acid for at least 15 min. Ca^{2+} imaging was performed on an upright confocal laser-scanning microscope (Olympus Fluoview 4000) equipped with a LUMPlanFl 60 $\times$ /0.90 W objective. GCaMP8f or OGB-1 were excited at 488 nm using 10% and 5% laser power, respectively, and emission was collected through a maximized pinhole using a grating and slit set to 510–550 nm. mScarlet was excited at 561 nm (2% power) and emission collected at 570–620 nm. Images were acquired with a 2 $\times$ digital zoom (190 nm/pixel) over a 512 $\times$ 48 pixel frame at 3.295 msec per frame. For each recording, 1 sec of baseline was collected prior to nerve stimulation, followed by 14 sec of imaging to capture 10 isolated single AP responses and a subsequent train of 17 APs. Nerves innervating segment 4 were stimulated using the silver wire in the suction pipette as the cathode and a bath wire as the anode, driven by an isolated pulse stimulator (DS2A, Digitimer) controlled by a Master-8 (A.M.P.I.). Stimuli were 1.5 V, 0.3 msec pulses delivered as 10 APs at 1 Hz, followed after a 2 sec delay by 17 APs at 21.3 Hz. Only preparations with stable baseline fluorescence and reproducible responses were analyzed. For each indicator, 7 larvae were imaged, with the full stimulus trial (10 @ 1 Hz followed by 17 @ 21.3 Hz) repeated three times per larva. Fluorescence was quantified from ROIs surrounding 1–2 non-terminating MN6/7-Ib boutons; background-subtracted stimulus fluorescence (F_{stim}) was obtained using cellSens FV software (v4.1.1, Evident Scientific) and further processed in Microsoft Excel. Bleaching (typically ~8 % and 12 % per 10 sec for GCaMP8f and OGB-1, respectively) was corrected by fitting a linear trend to the average pre-stimulus fluorescence in each 100 msec window preceding the 10 APs at 1 Hz and adjusting each F_{stim} according to its time and the slope of this

line. $\Delta F/F$ was then calculated as $(F_{\text{stim}} - F_{\text{rest}})/F_{\text{rest}}$, where F_{rest} is the mean fluorescence during the 100 msec preceding the first AP. mScarlet fluorescence was not used when benchmarking Scar8f against OGB-1 because a red dextran fluorophore was not co-loaded in constant proportion with OGB-1. For single-AP analysis, three trials per preparation were averaged, and the 10 responses were aligned so that the first rise in F_{stim} occurred 1.648 msec (half a sampling interval) after the stimulus. Amplitude was obtained from the intercept at time 0 of a single-exponential fit ($\Delta F/F = \Delta F/F_0 e^{-bt}$) to data from 14.828–499.193 msec (147 points), and the decay time constant τ was calculated as $1000 \text{ ms}/-b$. For train responses (17 @ 21.3 Hz), trials were averaged after alignment to the first rise in F_{stim} ; amplitude was taken as the peak $\Delta F/F$ after the final AP, the facilitation index was computed as the ratio of the final to the first response, and τ was derived from a single-exponential fit to 1 sec (304 points) of data following the peak.

Widefield Ca^{2+} imaging

Widefield Ca^{2+} imaging was conducted on muscle 4 NMJs using a Zeiss Axio Examiner A1 upright fixed-stage microscope equipped with a pco.panda 4.2 sCMOS camera (Excelitas) in 1.8 mM Ca^{2+} HL3 saline. High-power 470 nm LED (Thorlabs, M470L4) was used for illumination with a 63 $\times$ /1.0 NA water-dipping lens, with image acquisition controlled using Nikon NIS-Elements software. Regions of interest (ROIs) were defined to encompass terminal boutons of MN-Ib branches on muscle 4. Images were acquired at 256 $\times$ 256 pixels with a frame rate of 100 fps. GCaMP was imaged using a GFP filter cube (LED excitation $\sim$ 470/40 nm, emission $\sim$ 525/50 nm). Time-series imaging was performed for 60 secs without external stimulation to capture spontaneous quantal Ca^{2+} events. Under these widefield imaging conditions, SynapGCaMP signals exhibited only modest photobleaching, typically a $\sim$ 2–5% decrease in baseline fluorescence over a 60 sec acquisition. The modest bleaching was either corrected using the “bleaching correction” function in Huygens SVI or accounted for during $\Delta F/F$ quantification in the CaFire analysis pipeline using a sliding-baseline procedure (see CaFire methods). For simultaneous electrophysiological recordings, Clampex software (Molecular Devices) was employed to perform current-clamp recordings in gap-free mode. A digital TTL trigger signal from Clampex synchronized the imaging acquisition in NIS-Elements, ensuring precise temporal alignment between electrophysiological and imaging data. Data was collected from a minimum of six biological replicates per genotype.

STED imaging

STED super-resolution microscopy was performed as previously described (He et al., 2023 [Chien et al., 2025](#)). Briefly, STED imaging was performed with an Abberior STEDYCON system mounted on a Nikon Eclipse FN1 upright microscope. The system is equipped with four excitation lasers (640, 561, 488 and 405 nm), a pulsed STED laser at 775 nm, and three avalanche photodiode detectors operating in a single photon counting mode.

Alexa Fluor 488-conjugated anti-HRP was used to locate NMJs, whereas Abberior STAR Red and Alexa Fluor 594 secondary dyes were used for the STED channels. Depletion was performed at 775 nm with time-gated detection set to open from 1 ns to 7 ns after each excitation pulse. Emission was detected with two avalanche photodiodes (band-pass filters: $600 \pm 25 \text{ nm}$ for STAR RED and $675 \pm 25 \text{ nm}$ for Alexa Fluor 594). Images were acquired sequentially with a pixel dwell time of 10 μ s and 3 $\times$ line accumulation. Multichannel z-stack STED images were acquired using a 100 $\times$ Nikon Plan APO 1.45 NA oil immersion objective with a 20-nm fixed pixel size at 130 nm steps, which yields an effective lateral resolution of $\sim$ 60 nm. Each image covered one to two boutons (25–64 μm^2) on muscle 4 of segment A3. Raw STED images were corrected for thermal drift and channel crosstalk, then deconvolved in SVI Huygens software Essential 24.04.0 (Scientific Volume Imaging B.V.) using theoretical STED point-spread and default iteration settings of the Good’s MLE algorithm.

CaFire program

CaFire is a software tool designed for the analysis and processing of Ca^{2+} imaging data, compatible with both raw intensity data and pre-normalized signals (e.g., $\Delta F/F$ or $\Delta R/R$). The graphical user interface (GUI) of CaFire is implemented with the Python Tkinter library, while the visualization panel is powered by Matplotlib. The method of automated peak detection utilizes signal processing functions from SciPy (Virtanen et al., 2020). We define SNR operationally as the fractional fluorescence change ($\Delta F/F$). Peaks that are not detected automatically can be selected manually. CaFire identifies the nearest peak to the location where the user clicks. For peak quantification analysis, CaFire calculates rise and decay properties by performing curve fitting on data preceding and following the peak, respectively. The curve fitting is performed using the Levenberg–Marquardt gradient descent algorithm, implemented in the curve fit function of SciPy (Virtanen et al., 2020). The rise time constant is calculated by applying an exponential fit to the rise phase (Shemesh et al., 2020). The decay is modeled using a natural exponential decay function: $y = y_{\text{peak}} \cdot e^{-t/\tau}$ (Fleming et al., 2021).

After motion correction/processing, ROIs were drawn in Fiji and mean-intensity time series were exported as numeric tables. The traces were organized in an Excel (.xlsx) file, with the first row containing column headers (text labels) and all subsequent rows containing numeric data only. CaFire directly imports .xlsx tables and requires at least two columns per trace: time (or frame number) on the x-axis and intensity on the y-axis (additional ROIs can be placed in separate columns with distinct headers). If only frame numbers are provided, rise and decay tau results need to be converted to time separately by multiplying the fitted tau (in frames) by the frame interval ($\Delta t = 1000$ msec/frame rate). CaFire supports both single- and dual-channel imaging: for all analyses, GCaMP traces are imported as Channel #1 (signal; required), and for dual-channel datasets the mScarlet reference is additionally imported as Channel #2 (baseline/reference; optional). For single-channel recordings, CaFire can either accept user-supplied $\Delta F/F$ traces or compute $\Delta F/F$ from raw fluorescence: in the latter case, CaFire first estimates a time-varying baseline $F_0(t)$ using a sliding-window percentile method, with the window length, percentile, and time range user-configurable in the GUI (unless otherwise noted, we used a 101-point window, i.e., 50 frames before and after each time point, and the 30th percentile of values within that window to define F_0). Fluorescence signals are then expressed as $\Delta F/F(t) = [F(t) - F_0(t)] / F_0(t)$, which we report throughout the manuscript as an operational measure of signal-to-noise ratio (SNR). For ratiometric datasets, CaFire allows users to analyze either precomputed $\Delta R/R$ traces or to generate $\Delta R/R$ from raw two-channel fluorescence data. ROIs are drawn in Fiji/ImageJ and mean fluorescence time courses are extracted for the green GCaMP channel and the red mScarlet channel, yielding $F_{\text{GCaMP}}(t)$ and $F_{\text{mScarlet}}(t)$, which are imported into CaFire as the signal and reference channels, respectively. If desired, the mScarlet signal can be smoothed in CaFire using a user-specified moving-average window (e.g., 8 frames at ~ 118 fps) to reduce high-frequency noise without altering the underlying ratio dynamics. In $\Delta R/R$ mode, CaFire computes a per-frame ratio $R(t) = F_{\text{GCaMP}}(t) / F_{\text{mScarlet}}(t)$ and estimates a baseline ratio $R_0(t)$ using the same sliding-window percentile procedure; for Bar8m active-zone analysis, we used a 101-point window and the lowest 10% of ratio values within each window to define R_0 . Ratiometric responses are then reported as $\Delta R/R(t) = [R(t) - R_0(t)] / R_0(t)$. CaFire can plot raw, $\Delta F/F$, or ratiometric traces, and $F_0(t)$ or $R_0(t)$ can be visualized in the raw-data view for quality control. The source code and a Windows executable, together with additional instructions (including ImageJ workflow instructions, an Excel template, CaFire loading screenshots, and a brief step-by-step usage guide), are available for download on CaFire's GitHub repository (<https://github.com/linj7/CaFire>) and has been archived with a DOI at Zenodo: 10.5281/zenodo.1552996. The software is released under the MIT License.

Statistical analysis

Data were analyzed using NIS Elements software (Nikon), MiniAnalysis (Synaptosoft), SVI Hugyens Essential (version 22.10), CaFire (<https://github.com/linj7/CaFire>), GraphPad Prism (version 10.0), and Microsoft Excel software (version 16.22). For comparisons between two groups, we used unpaired two-tailed Student's t-tests. For comparisons among more than two groups, we used one-

way ANOVA followed by Tukey's multiple comparisons test. Cumulative distributions of mEPSP amplitudes and Ca^{2+} mini amplitudes were compared using two-sample Kolmogorov-Smirnov tests. Pearson correlation coefficients and linear regression were used to assess relationships between optical and electrophysiological miniature event amplitudes, as well as between AZ size and Ca^{2+} signal amplitude. When outlier analysis was required (e.g., analyzing AZs), outliers were identified using the interquartile range (IQR) method. To quantify AZ-to-AZ variability, we performed paired two-tailed t-tests between every pair of active zones within the same preparation, comparing peak responses across stimuli. For Bar8m line-scan recordings, ~79% of AZ pairs were significantly different at $\alpha < 0.05$, whereas for Bar8f area-scan recordings only ~8% of AZ pairs were significant, indicating substantially greater AZ-to-AZ diversity when measured at high temporal resolution. For visual marking of outlying AZs in the scatter plots, we computed the interquartile range (IQR; Q1–Q3) from the individual responses pooled across AZs (15 stimuli per AZ, 30 AZs per genotype) and highlighted AZs whose mean response lay above Q3 or below Q1. The IQR was used solely as a robust measure of dispersion for visualization, and not as a formal hypothesis test. The fraction of significant AZ pairs is reported on the corresponding plots. Throughout the manuscript, n refers to the number of animals (biological replicates). All Ca^{2+} imaging and electrophysiological recordings were performed at abdominal muscle 6, segment A3 of third-instar larvae. For imaging experiments, each larva contributed 1–2 NMJs in total; at each NMJ we typically imaged 2–3 terminal boutons and acquired 2–3 time-series recordings from different boutons. For the standard stimulation protocol, 1-msec pulses were delivered at 1 Hz during a 15-sec imaging epoch (yielding ~14–15 stimuli per trial). In all figures, error bars indicate $\pm$ SEM, with the following statistical significance: $p < 0.05$ (*), $p < 0.01$ (**), $p < 0.001$ (***), $p < 0.0001$ (****); ns=not significant. Additional statistical details for all experiments are summarized in Table S1.

Data availability

All acquired data are available upon request to the lead contact. All fly stocks and molecular constructs generated in this study will be shared upon request. We are also in the process of depositing newly generated Scar8f/m, Bar8f/m, and SynapGCaMP sensors to the Bloomington Drosophila Stock Center for public dissemination. The source code for the CaFire analysis software is publicly available on GitHub at <https://github.com/linj7/CaFire> and has been archived with a DOI at Zenodo: 10.5281/zenodo.1552996.

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Additional information

Author contributions

J.C. and D.D. designed the research; J.L. developed the CaFire software; J.C., J.L., K.H., L.W., Y.H., C.Q., J.W., C.D., and G.M. performed experiments and analyzed the data. The manuscript was written by J.C. and D.D. with feedback from the other authors.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Google Gemini Advanced 2.5 Pro to line edit an early version of the manuscript to improve clarity and readability. After using this tool, the authors subsequently reviewed and edited the content and take full responsibility for the content of the publication.

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Additional files

[Supplemental Table 1](#) 

[Supplemental Table 2](#) 

[Supplemental Video 1](#) 

[Supplemental Figure 1](#) 

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Peer reviews

Reviewer #1 (Public review):

Summary:

Chen et al. engineered and characterized a suite of next-generation GECIs for the *Drosophila* NMJ that allow for the visualization of calcium dynamics within the presynaptic compartment, at presynaptic active zones, and in the postsynaptic compartment. These GECIs include ratiometric presynaptic Scar8m (targeted to synaptic vesicles), ratiometric active zone localized Bar8f (targeted to the scaffold molecule BRP), and postsynaptic

SynapGCaMP8m. The authors demonstrate that these new indicators are a large improvement on the widely used GCaMP6 and GCaMP7 series GECIs, with increased speed and sensitivity. They show that presynaptic Scar8m accurately captures presynaptic calcium dynamics with superior sensitivity to the GCaMP6 and GCaMP7 series and with similar kinetics to chemical dyes. The active-zone targeted Bar8f sensor was assessed for the ability to detect release-site specific nanodomain changes, but the authors concluded that this sensor is still too slow to accurately do so. Lastly, the use of postsynaptic SynapGCaMP8m was shown to enable the detection of quantal events with similar resolution to electrophysiological recordings. Finally, the authors developed a Python-based analysis software, CaFire, that enables automated quantification of evoked and spontaneous calcium signals. These tools will greatly expand our ability to detect activity at individual synapses without the need for chemical dyes or electrophysiology.

Strengths:

In this study, the authors rigorously compare their newly engineered GECIs to those previously used at the *Drosophila* NMJ, highlighting improvements in localization, speed, and sensitivity. These comparisons appropriately substantiate the authors claim that their GECIs are superior to the ones currently in use.

The authors demonstrate the ability of Scar8m to capture subtle changes in presynaptic calcium resulting from differences between MN-Ib and MN-Is terminals and from the induction of presynaptic homeostatic potentiation (PHP), rivaling the sensitivity of chemical dyes.

The improved postsynaptic SynapGCaMP8m is shown to approach the resolution of electrophysiology in resolving quantal events.

The authors created a publicly available pipeline that streamlines and standardizes analysis of calcium imaging data.

<https://doi.org/10.7554/eLife.107939.2.sa3>

Reviewer #2 (Public review):

Summary:

Calcium ions play a key role in synaptic transmission and plasticity. To improve calcium measurements at synaptic terminals, previous studies have targeted genetically encoded calcium indicators (GECIs) to pre- and postsynaptic locations. Here, Chen et al. improve these constructs by incorporating the latest GCaMP8 sensors and a stable red fluorescent protein to enable ratiometric measurements. Extensive characterization in the *Drosophila* neuromuscular junction demonstrates favorable performance of these new constructs relative to previous genetically encoded and synthetic calcium indicators in reporting synaptic calcium events. In addition, they develop a new analysis platform, 'CaFire', to facilitate automated quantification. Impressively, by positioning postsynaptic GCaMP8m near glutamate receptors, the authors show that their sensors can report miniature synaptic events with speed and sensitivity approaching that of intracellular electrophysiological recordings. These new sensors and the analysis platform provide a valuable tool for resolving synaptic events using all-optical methods.

Strength:

The authors present rigorous characterization of their sensors using well-established assays. They employ immunostaining and super-resolution STED microscopy to confirm correct subcellular targeting. Additionally, they quantify response amplitude, rise and decay kinetics, and provide side-by-side comparisons with earlier-generation GECIs and synthetic dyes.

Importantly, they show that the new sensors can reproduce known differences in evoked Ca^{2+} responses between distinct nerve terminals. Finally, they present what appears to be the first simultaneous calcium imaging and intracellular mEPSP recording to directly assess the sensitivity of different sensors in detecting individual miniature synaptic events.

The revised version contains extensive new data and clarification that fully addressed my previous concerns. In particular, I appreciate the side-by-side comparison with synthetic calcium indicator OGB-1 and the cytosolic version of GCaMP8m (now presented in Figure 3), which compellingly supports the favorable performance of their new sensors.

Weakness:

I have only one remaining suggestion about the precision of terminology, which I do think is important. The authors clarified in the revision that they "define SNR operationally as the fractional fluorescence change ($\Delta F/F$).", and basically present $\Delta F/F$ values whenever they mentioned about SNR. However, if the intention is to present $\Delta F/F$ comparisons, I would strongly suggest replacing all mentions of "SNR" in the manuscript with " $\Delta F/F$ " or "fractional/relative fluorescence change".

SNR and $\Delta F/F$ are fundamentally different quantities, each with a well-defined and distinct meaning: SNR measures fluorescence change relative to baseline fluctuations (noise), whereas $\Delta F/F$ measures fluorescence change relative to baseline fluorescence (F_0). While larger $\Delta F/F$ values often correlate with improved detectability, SNR also depends on additional factors such as indicator brightness, light collection efficiency, camera noise, and the stability of the experimental preparation. Referring to $\Delta F/F$ as SNR can therefore be misleading and may cause confusion for readers, particularly those from quantitative imaging backgrounds. Clarifying the terminology by consistently using $\Delta F/F$ would improve conceptual accuracy without requiring any reanalysis of the data.

<https://doi.org/10.7554/eLife.107939.2.sa2>

Reviewer #3 (Public review):

Genetically encoded calcium indicators (GECIs) are essential tools in neurobiology and physiology. Technological constraints in targeting and kinetics of previous versions of GECIs have limited their application at the subcellular level. Chen et al. present a set of novel tools that overcome many of these limitations. Through systematic testing in the *Drosophila* NMJ, they demonstrate improved targeting of GCaMP variants to synaptic compartments and report enhanced brightness and temporal fidelity using members of the GCaMP8 series. These advancements are likely to facilitate more precise investigation of synaptic physiology. This manuscript could be improved by further testing the GECIs across physiologically relevant ranges of activity, including at high frequency and over long imaging sessions. Moreover, while the authors provide a custom software package (CaFire) for Ca^{2+} imaging analysis, comparisons to existing tools and more guidance for broader usability are needed.

In this revised version, Chen et al. answered most of our concerns. The tools developed here will be useful for the community.

<https://doi.org/10.7554/eLife.107939.2.sa1>

Author response:

The following is the authors' response to the original reviews.

| **Reviewer #1**

Chen et al. engineered and characterized a suite of next-generation GECIs for the Drosophila NMJ that allow for the visualization of calcium dynamics within the presynaptic compartment, at presynaptic active zones, and in the postsynaptic compartment. These GECIs include ratiometric presynaptic Scar8m (targeted to synaptic vesicles), ratiometric active zone localized Bar8f (targeted to the scaffold molecule BRP), and postsynaptic SynapGCaMP8m. The authors demonstrate that these new indicators are a large improvement on the widely used GCaMP6 and GCaMP7 series GECIs, with increased speed and sensitivity. They show that presynaptic Scar8m accurately captures presynaptic calcium dynamics with superior sensitivity to the GCaMP6 and GCaMP7 series and with similar kinetics to chemical dyes. The active-zone targeted Bar8f sensor was assessed for the ability to detect release-site-specific nanodomain changes, but the authors concluded that this sensor is still too slow to accurately do so. Lastly, the use of postsynaptic SynapGCaMP8m was shown to enable the detection of quantal events with similar resolution to electrophysiological recordings. Finally, the authors developed a Python-based analysis software, CaFire, that enables automated quantification of evoked and spontaneous calcium signals. These tools will greatly expand our ability to detect activity at individual synapses without the need for chemical dyes or electrophysiology.

We thank this Reviewer for the overall positive assessment of our manuscript and for the incisive comments.

(1) The role of Excel in the pipeline could be more clearly explained. Lines 182-187 could be better worded to indicate that CaFire provides analysis downstream of intensity detection in ImageJ. Moreover, the data type of the exported data, such as .csv or .xlsx, should be indicated instead of 'export to graphical program such as Microsoft Excel'.

We thank the Reviewer for these comments, many of which were shared by the other reviewers. In response, we have now 1) more clearly explained the role of Excel in the CaFire pipeline (lines 677-681), 2) revised the wording in lines 676-679 to indicate that CaFire provides analysis downstream of intensity detection in ImageJ, and 3) Clarified the exported data type to Excel (lines 677-681). These efforts have improved the clarity and readability of the CaFire analysis pipeline.

(2) In Figure 2A, the 'Excel' step should either be deleted or included as 'data validation' as ImageJ exports don't require MS Excel or any specific software to be analysed. (Also, the graphic used to depict Excel software in Figure 2A is confusing.)

We thank the reviewer for this helpful suggestion. In the Fig. 2A, we have changed the Excel portion and clarified the processing steps in the revised methods. Specifically, we now indicate that ROIs are first selected in Fiji/ImageJ and analyzed to obtain time-series data containing both the time information and the corresponding imaging mean intensity values. These data are then exported to a spreadsheet file (e.g., Excel), which is used to organize the output before being imported into CaFire for subsequent analysis. These changes can be found in the Fig. 2A and methods (lines 676-681).

(3) Figure 2B should include the 'Partition Specification' window (as shown on the GitHub) as well as the threshold selection to give the readers a better understanding of how the tool works.

We absolutely agree with this comment, and have made the suggested changes to the Fig. 2B. In particular, we have replaced the software interface panels and now include windows illustrating the Load File, Peak Detection, and Partition functions. These updated screenshots provide a clearer view of how CaFire is used to load the data, detect events, and perform partition specification for subsequent analysis. We agree these changes will give the readers a better understanding of how the tool works, and we thank the reviewer for this comment.

(4) The presentation of data is well organized throughout the paper. However, in Figure 6C, it is unclear how the heatmaps represent the spatiotemporal fluorescence dynamics of each indicator. Does the signal correspond to a line drawn across the ROI shown in Figure 6B? If so, this should be indicated.

We apologize that the heatmaps were unclear in Fig panel 6C (Fig. 7C in the Current revision). Each heatmap is derived from a one-pixel-wide vertical line within a miniature-event ROI. These heatmaps correspond to the fluorescence change in the indicated SynapGCaMP variant of individual quantal events and their traces shown in Fig. 7C, with a representative image of the baseline and peak fluorescence shown in Fig. 7B. Specifically, we have added the following to the revised Fig. 7C legend:

The corresponding heatmaps below were generated from a single vertical line extracted from a representative miniature-event ROI, and visualize the spatiotemporal fluorescence dynamics ($\Delta F/F$) along that line over time.

(5) In Figure 6D, the addition of non-matched electrophysiology recordings is confusing. Maybe add "at different time points" to the end of the 6D legend, or consider removing the electrophysiology trace from Figure 6D and referring the reader to the traces in Figure 7A for comparison (considering the same point is made more rigorously in Figure 7).

This is a good point, one shared with another reviewer. We apologize this was not clear, and have now revised this part of the figure to remove the electrophysiological traces in what is now Fig. 7 while keeping the paired ones still in what is now Fig. 8A as suggested by the reviewer. We agree this helps to clarify the quantal calcium transients.

(6) In GitHub, an example ImageJ Script for analyzing the images and creating the inputs for CaFire would be helpful to ensure formatting compatibility, especially given potential variability when exporting intensity information for two channels. In the Usage Guide, more information would be helpful, such as how to select $\Delta R/R$, ideally with screenshots of the application being used to analyze example data for both single-channel and two-channel images.

We agree that additional details added to the GitHub would be helpful for users of CaFire. In response, we have now added the following improvements to the GitHub site:

- ImageJ operation screenshots

Step-by-step illustrations of ROI drawing and Multi Measure extraction.

- Example Excel file with time and intensity values

Demonstrates the required data format for CaFire import, including proper headers.

- CaFire loading screenshots for single-channel and dual-channel imaging

Shows how to import GCaMP into Channel 1 and mScarlet into Channel 2.

- Peak Detection and Partition setting screenshots

Visual examples of automatic peak detection, manual correction, and trace partitioning.

- Instructions for ROI Extraction and CaFire Analysis

A written guide describing the full workflow from ROI selection to CaFire data export.

These changes have improved the usability and accessibility of CaFire, and we thank the reviewer for these points.

Reviewer #2

Calcium ions play a key role in synaptic transmission and plasticity. To improve calcium measurements at synaptic terminals, previous studies have targeted genetically encoded calcium indicators (GECIs) to pre- and postsynaptic locations. Here, Chen et al. improve these constructs by incorporating the latest GCaMP8 sensors and a stable red fluorescent protein to enable ratiometric measurements. In addition, they develop a new analysis platform, 'CaFire', to facilitate automated quantification. Using these tools, the authors demonstrate favorable properties of their sensors relative to earlier constructs. Impressively, by positioning postsynaptic GCaMP8m near glutamate receptors, they show that their sensors can report miniature synaptic events with speed and sensitivity approaching that of intracellular electrophysiological recordings. These new sensors and the analysis platform provide a valuable tool for resolving synaptic events using all-optical methods.

We thank the Reviewer for their overall positive evaluation and comments.

Major comments:

(1) While the authors rigorously compared the response amplitude, rise, and decay kinetics of several sensors, key parameters like brightness and photobleaching rates are not reported. I feel that including this information is important as synaptically tethered sensors, compared to freely diffusible cytosolic indicators, can be especially prone to photobleaching, particularly under the high-intensity illumination and high-magnification conditions required for synaptic imaging. Quantifying baseline brightness and photobleaching rates would add valuable information for researchers intending to adopt these tools, especially in the context of prolonged or high-speed imaging experiments.

This is a good point made by the reviewer, and one we agree will be useful for researchers to be aware. First, it is important to note that the photobleaching and brightness of the sensors will vary depending on the nature of the user's imaging equipment, which can vary significantly between widefield microscopes (with various LED or halogen light sources for illumination), laser scanning systems (e.g., line scans with confocal systems), or area scanning systems using resonant scanners (as we use in our current study). Under the same imaging settings, GCaMP8f and 8m exhibit comparable baseline fluorescence, whereas GCaMP6f and 6s are noticeably dimmer; because our aim is to assess each reagent's potential under optimal conditions, we routinely adjust excitation/camera parameters before acquisition to place baseline fluorescence in an appropriate dynamic range. As an important addition to this study, motivated by the reviewer's comments above, we now directly compare neuronal cytosolic GCaMP8m expression with our Scar8m sensor, showing higher sensitivity with Scar8m (now shown in the new Fig. 3F-H).

Regarding photobleaching, GCaMP signals are generally stable, while mScarlet is more prone to bleaching: in presynaptic area scanned confocal recordings, the mScarlet channel drops by ~15% over 15 secs, whereas GCaMP6s/8f/8m show no obvious bleaching over the same window (lines 549-553). In contrast, presynaptic widefield imaging using an LED system (CCD), GCaMP8f shows ~8% loss over 15 secs (lines 610-611). Similarly, for postsynaptic SynapGCaMP6f/8f/8m, confocal resonant area scans show no obvious bleaching over 60 secs, while widefield shows ~2–5% bleaching over 60 secs (lines 634-638). Finally, in active-zone/BRP calcium imaging (confocal), mScarlet again bleaches by ~15% over 15 s, while GCaMP8f/8m show no obvious bleaching. The mScarlet-channel bleaching can be corrected in Huygens SVI (Bleaching correction or via the Deconvolution Wizard), whereas we avoid

applying bleaching correction to the green GCaMP channel when no clear decay is present to prevent introducing artifacts. This information is now added to the methods (lines 548-553).

(2) In several places, the authors compare the performance of their sensors with synthetic calcium dyes, but these comparisons are based on literature values rather than on side-by-side measurements in the same preparation. Given differences in imaging conditions across studies (e.g., illumination, camera sensitivity, and noise), parameters like indicator brightness, SNR, and photobleaching are difficult to compare meaningfully. Additionally, the limited frame rate used in the present study may preclude accurate assessment of rise times relative to fast chemical dyes. These issues weaken the claim made in the abstract that "...a ratiometric presynaptic GCaMP8m sensor accurately captures .. Ca^{2+} changes with superior sensitivity and similar kinetics compared to chemical dyes." The authors should clearly acknowledge these limitations and soften their conclusions. A direct comparison in the same system, if feasible, would greatly strengthen the manuscript.

We absolutely agree with these points made the reviewer, and have made a concerted effort to address them through the following:

We have now directly compared presynaptic calcium responses on the same imaging system using the chemical dye Oregon Green Bapta-1 (OGB-1), one of the primary synthetic calcium indicators used in our field. These experiments reveal that Scar8f exhibits markedly faster kinetics and an improved signal-to-noise ratio compared to OGB-1, with higher peak fluorescence responses (Scar8f: 0.32, OGB-1: 0.23). The rise time constants of the two indicators are comparable (both ~3 msec), whereas the decay of Scar8f is faster than that of OGB-1 (Scar8f: ~40, OGB-1: ~60), indicating more rapid signal recovery. These results now directly demonstrate the superiority of the new GCaMP8 sensors we have engineered over conventional synthetic dyes, and are now presented in the new Fig. 3A-E of the manuscript.

We agree with the reviewer that, in the original submission, the relatively slow resonant area scans (~115 fps) limited the temporal resolution of our rise time measurements. To address this, we have re-measured the rise time using higher frame-rate line scans (kHz). For Scar8f, the rise time constant was 6.736 msec at ~115 fps resonant area scanned, but shortened to 2.893 msec when imaged at ~303 fps, indicating that the original protocol underestimated the true kinetics. In addition, for Bar8m, area scans at ~118 fps yielded a rise time constant of 9.019 msec, whereas line scans at ~1085 fps reduced the rise time constant to 3.230 msec. These new measurements are now incorporated into the manuscript (Figs. 3,4, and 6) to more accurately reflect the fast kinetics of these indicators.

(3) The authors state that their indicators can now achieve measurements previously attainable with chemical dyes and electrophysiology. I encourage the authors to also consider how their tools might enable new measurements beyond what these traditional techniques allow. For example, while electrophysiology can detect summed mEPSPs across synapses, imaging could go a step further by spatially resolving the synaptic origin of individual mEPSP events. One could, for instance, image MN-Ib and MN-Is simultaneously without silencing either input, and detect mEPSP events specific to each synapse. This would enable synapse-specific mapping of quantal events - something electrophysiology alone cannot provide. Demonstrating even a proof-of-principle along these lines could highlight the unique advantages of the new tools by showing that they not only match previous methods but also enable new types of measurements.

These are excellent points raised by the reviewer. In response, we have done the following:

We have now included a supplemental video as "proof-of-principle" data showing simultaneous imaging of SynapGCaMP8m quantal events at both MN-Is and -Ib,

demonstrating that synapse-specific spatial mapping of quantal events can be obtained with this tool (see new Supplemental Video 1).

We have also included an additional discussion of the potential and limitations of these tools for new measurements beyond conventional approaches. This discussion is now presented in lines 419-421 in the manuscript.

(4) For ratiometric measurements, it is important to estimate and subtract background signals in each channel. Without this correction, the computed ratio may be skewed, as background adds an offset to both channels and can distort the ratio. However, it is not clear from the Methods section whether, or how, background fluorescence was measured and subtracted.

This is a good point, and we agree more clarification about how ratiometric measurements were made is needed. In response, we have now added the following to the Methods section (lines 548-568):

Time-lapse videos were stabilized and bleach-corrected prior to analysis, which visibly reduced frame-to-frame motion and intensity drift. In the presynaptic and active-zone mScarlet channel, a bleaching factor of ~1.15 was observed during the 15 sec recording. This bleaching can be corrected using the “Bleaching correction” tool in Huygens SVI. For presynaptic and active-zone GCaMP signals, there was minimal bleaching over these short imaging periods. Therefore, the bleaching correction step for GCaMP was skipped. Both GCaMP and mScarlet channels were processed using the default settings in the Huygens SVI “Deconvolution Wizard” (with the exception of the bleaching correction option). Deconvolution was performed using the CMLE algorithm with the Huygens default stopping criterion and a maximum of 30 iterations, such that the algorithm either converged earlier or, if convergence was not reached, was terminated at this 30iteration limit; no other iteration settings were used across the GCaMP series. ROIs were drawn on the processed images using Fiji ImageJ software, and mean fluorescence time courses were extracted for the GCaMP and mScarlet channels, yielding $F_{\text{GCaMP}}(t)$ and $F_{\text{mScarlet}}(t)$. $F(t)$ s were imported into CaFire with GCaMP assigned to Channel #1 (signal; required) and mScarlet to Channel #2 (baseline/reference; optional). If desired, the mScarlet signal could be smoothed in CaFire using a user-specified moving-average window to reduce high-frequency noise. In CaFire’s $\Delta R/R$ mode, the per-frame ratio was computed as $R(t) = F_{\text{GCaMP}}(t) / F_{\text{mScarlet}}(t)$; a baseline ratio R_0 was estimated from the pre-stimulus period, and the final response was reported as $\Delta R/R(t) = [R(t) - R_0] / R_0$, which normalizes GCaMP signals to the co-expressed mScarlet reference and thereby reduces variability arising from differences in sensor expression level or illumination across AZs.

(5) At line 212, the authors claim "... GCaMP8m showing 345.7% higher SNR over GCaMP6s....(Fig. 3D and E) ", yet the cited figure panels do not present any SNR quantification. Figures 3D and E only show response amplitudes and kinetics, which are distinct from SNR. The methods section also does not describe details for how SNR was defined or computed.

This is another good point. We define SNR operationally as the fractional fluorescence change ($\Delta F/F$). Traces were processed with CaFire, which estimates a per-frame baseline $F_0(t)$ with a user-configurable sliding window and percentile. In the Load File panel, users can specify both the length of the moving baseline window and the desired percentile; the default settings are a 50-point window and the 30th percentile, representing a 101-point window centered on each time point (previous 50 to next 50 samples) and took the lower 30% of values within that window to estimate $F_0(t)$. The signal was then computed as $\Delta F/F = [F(t) - F_0(t)] / F_0(t)$. This $\Delta F/F$ value is what we report as SNR throughout the manuscript and is now discussed explicitly in the revised methods (lines 686-693).

(6) Lines 285-287 "As expected, summed ΔF values scaled strongly and positively with AZ size (Fig. 5F), reflecting a greater number of Cav2 channels at larger AZs". I am not sure about this conclusion. A positive correlation between summed ΔF values and AZ size could simply reflect more GCaMP molecules in larger AZs, which would give rise to larger total fluorescence change even at a given level of calcium increase.

The reviewer makes a good point, one that we agree should be clarified. The reviewer is indeed correct that larger active zones should have more abundant BRP protein, which in turn will lead to a higher abundance of the Bar8f sensor, which should lead to a higher GCaMP response simply by having more of this sensor. However, the inclusion of the ratiometric mScarlet protein should normalize the response accurately, correcting for this confound, in which the higher abundance of GCaMP should be offset (normalized) by the equally (stoichiometric) higher abundance of mScarlet. Therefore, when the $\Delta R/R$ is calculated, the differences in GCaMP abundance at each AZ should be corrected for the ratiometric analysis. We now use an improved BRP::mScarlet3::GCaMP8m (Bar8m) and compute $\Delta R/R$ with $R(t) = F_{\text{GCaMP8m}}/F_{\text{mScarlet3}}$. ROIs were drawn over individual AZs (Fig. 6B). CaFire estimated R_0 with a sliding 101-point window using the lowest 10% of values, and responses were reported as $\Delta R/R = [R - R_0]/R_0$. Area-scan examples (118 fps) show robust $\Delta R/R$ transients (peaks ≈ 1.90 and 3.28 ; tau rise ≈ 9.0 – 9.3 ms; Fig. 6C, middle).

We have now made these points more clearly in the manuscript (lines 700-704) and moved the Bar8f intensity vs active zone size data to Table S1. Together, these revisions improve the indicator-abundance confound (via mScarlet normalization).

(6) Lines 313-314: "SynapGCaMP quantal signals appeared to qualitatively reflect the same events measured with electrophysiological recordings (Fig. 6D)." This statement is quite confusing. In Figure 6D, the corresponding calcium and ephys traces look completely different and appear to reflect distinct sets of events. It was only after reading Figure 7 that I realized the traces shown in Figure 6D might not have been recorded simultaneously. The authors should clarify this point.

Yes, we absolutely agree with this point, one shared by Reviewer 1. In response, we have removed the electrophysiological traces in Fig. 6 to clarify that just the calcium responses are shown, and save the direct comparison for the Fig. 7 data (now revised Fig. 8).

(8) Lines 310-313: "SynapGCaMP8m striking an optimal balance between speed and sensitivity", and Lines 314-316: "We conclude that SynapGCaMP8m is an optimal indicator to measure quantal transmission events at the synapse." Statements like these are subjective. In the authors' own comparison, GCaMP8m is significantly slower than GCaMP8f (at least in terms of decay time), despite having a moderately higher response amplitude. It is therefore unclear why GCaMP8m is considered 'optimal'. The authors should clarify this point or explain their rationale for prioritizing response amplitude over speed in the context of their application.

This is another good point that we agree with, as the "optimal" sensor will of course depend on the user's objectives. Hence, we used the term "an optimal sensor" to indicate it is what we believed to be the best one for our own uses. However, this point should be clarified and better discussed. In response, we have revised the relevant sections of the manuscript to better define why we chose the 8m sensors to strike an optimal balance of speed and sensitivity for our uses, and go on to discuss situations in which other sensor variants might be better suited. These are now presented in lines 223-236 in the revised manuscript, and we thank the reviewer for making these comments, which have improved our study.

Minor comments

(1) Please include the following information in the Methods section:

(a) For Figures 3 and 4, specify how action potentials were evoked. What type of electrodes were used, where were they placed, and what amount of current or voltage was applied?

We apologize for neglecting to include this information in the original submission. We have now added this information to the revised Methods section (lines 537-543).

(b) For imaging experiments, provide information on the filter sets used for each imaging channel, and describe how acquisition was alternated or synchronized between the green and red channels in ratiometric measurements. Additionally, please report the typical illumination intensity (in mW/mm²) for each experimental condition.

We thank the reviewer for this helpful comment. We have now added detailed information about the imaging configuration to the Methods (lines 512-528) with the following:

Ca²⁺ imaging was conducted using a Nikon A1R resonant scanning confocal microscope equipped with a 60x/1.0 NA water-immersion objective (refractive index 1.33). GCaMP signals were acquired using the FITC/GFP channel (488-nm laser excitation; emission collected with a 525/50-nm band-pass filter), and mScarlet/mCherry signals were acquired using the TRITC/mCherry channel (561-nm laser excitation; emission collected with a 595/50-nm band-pass filter). ROIs focused on terminal boutons of MN-Ib or -Is motor neurons. For both channels, the confocal pinhole was set to a fixed diameter of 117.5 μ m (approximately three Airy units under these conditions), which increases signal collection while maintaining adequate optical sectioning. Images were acquired as 256 $\times$ 64 pixel frames (two 12-bit channels) using bidirectional resonant scanning at a frame rate of $\sim$ 118 frames/s; the scan zoom in NIS-Elements was adjusted so that this field of view encompassed the entire neuromuscular junction and was kept constant across experiments. In ratiometric recordings, the 488-nm (GCaMP) and 561-nm (mScarlet) channels were acquired in a sequential dual-channel mode using the same bidirectional resonant scan settings: for each time point, a frame was first collected in the green channel and then immediately in the red channel, introducing a small, fixed frame-to-frame temporal offset while preserving matched spatial sampling of the two channels.

Directly measuring the absolute laser power at the specimen plane (and thus reporting illumination intensity in mW/mm²) is technically challenging on this resonant-scanning system, because it would require inserting a power sensor into the beam path and perturbing the optical alignment; consequently, we are unable to provide reliable absolute mW/mm² values. Instead, we now report all relevant acquisition parameters (objective, numerical aperture, refractive index, pinhole size, scan format, frame rate, and fixed laser/detector settings) and note that laser powers were kept constant within each experimental series and chosen to minimize bleaching and phototoxicity while maintaining an adequate signal-to-noise ratio. We have now added the details requested in the revised Methods section (lines 512-535), including information about the filter sets, acquisition settings, and typical illumination intensity.

(2) Please clarify what the thin versus thick traces represent in Figures 3D, 3F, 4C, and 4E. Are the thin traces individual trials from the same experiment, or from different experiments/animals? Does the thick trace represent the mean/median across those trials, a fitted curve, or a representative example?

We apologize this was not more clear in the original submission. Thin traces are individual stimulus-evoked trials (“sweeps”) acquired sequentially from the same muscle/NMJ in a single preparation; the panel is shown as a representative example of recordings collected across animals. The thick colored trace is the trial-averaged waveform (arithmetic mean) of those thin traces after alignment to stimulus onset and baseline subtraction (no additional smoothing beyond what is stated in Methods). The thick black curve over the decay phase is a

single-exponential fit used to estimate τ . Specifically, we fit the decay segment by linear regression on the natural-log-transformed baseline-subtracted signal, which is equivalent to fitting $y = y_{peak} \cdot e^{-t/\tau_{decay}}$ over the decay window (revised Fig.4D and Fig.5C legends).

(3) Please clarify what the reported sample size (n) represents. Does it indicate the number of experimental repeats, the number of boutons or PSDs, or the number of animals?

Again, we apologize this was not clear. (n) refers to the number of animals (biological replicates), which is reported in Supplementary Table 1. All imaging was performed at muscle 6, abdominal segment A3. Per preparation, we imaged 1-2 NMJs in total, with each imaging targeting 2–3 terminal boutons at the target NMJ and acquired 2–3 imaging stacks choosing different terminal boutons per NMJ. For the standard stimulation protocol, we delivered 1 Hz stimulation for 1ms and captured 14 stimuli in a 15s time series imaging (lines 730-736).

Reviewer #3

Genetically encoded calcium indicators (GECIs) are essential tools in neurobiology and physiology. Technological constraints in targeting and kinetics of previous versions of GECIs have limited their application at the subcellular level. Chen et al. present a set of novel tools that overcome many of these limitations. Through systematic testing in the Drosophila NMJ, they demonstrate improved targeting of GCaMP variants to synaptic compartments and report enhanced brightness and temporal fidelity using members of the GCaMP8 series. These advancements are likely to facilitate more precise investigation of synaptic physiology.

This is a comprehensive and detailed manuscript that introduces and validates new GECI tools optimized for the study of neurotransmission and neuronal excitability. These tools are likely to be highly impactful across neuroscience subfields. The authors are commended for publicly sharing their imaging software.

This manuscript could be improved by further testing the GECIs across physiologically relevant ranges of activity, including at high frequency and over long imaging sessions. The authors provide a custom software package (CaFire) for Ca²⁺ imaging analysis; however, to improve clarity and utility for future users, we recommend providing references to existing Ca²⁺ imaging tools for context and elaborating on some conceptual and methodological aspects, with more guidance for broader usability. These enhancements would strengthen this already strong manuscript.

We thank the Reviewer for their overall positive evaluation and comments.

Major comments:

(1) *Evaluation of the performance of new GECI variants using physiologically relevant stimuli and frequency. The authors took initial steps towards this goal, but it would be helpful to determine the performance of the different GECIs at higher electrical stimulation frequencies (at least as high as 20 Hz) and for longer (10 seconds) (Newman et al, 2017). This will help scientists choose the right GECI for studies testing the reliability of synaptic transmission, which generally requires prolonged highfrequency stimulation.*

We appreciate this point by the reviewer and agree it would be of interest to evaluate sensor performance with higher frequency stimulation and for a longer duration. In response, we performed a variety of stimulation protocols at high intensities and times, but found the data to be difficult to separate individual responses given the decay kinetics of all calcium sensors. Hence, we elected not to include these in the revised manuscript. However, we have now included an evaluation of the sensors with 20 Hz electrical stimulation for ~1 sec using a

direct comparison of Scar8f with OGB-1. These data are now presented in a new Fig. 3D,E and discussed in the manuscript (lines 396-403).

(2) *CaFire*.

The authors mention, in line 182: 'Current approaches to analyze synaptic Ca²⁺ imaging data either repurpose software designed to analyze electrophysiological data or use custom software developed by groups for their own specific needs.' References should be provided. CaImAn comes to mind (Giovannucci et al., 2019, eLife), but we think there are other software programs aimed at analyzing Ca²⁺ imaging data that would permit such analysis.

Thank you for the thoughtful question. At this stage, we're unable to provide a direct comparison with existing analysis workflows. In surveying prior studies that analyze *Drosophila* NMJ Ca²⁺ imaging traces, we found that most groups preprocess images in Fiji/ImageJ and then rely on their own custom-made MATLAB or Python scripts for downstream analysis (see Blum et al. 2021; Xing and Wu 2018). Because these pipelines vary widely across labs, a standardized head-to-head evaluation isn't currently feasible. With CaFire, our goal is to offer a simple, accessible tool that does not require coding experience and minimizes variability introduced by custom scripts. We designed CaFire to lower the barrier to entry, promote reproducibility, and make quantal event analysis more consistent across users. We have added references to the sentence mentioned above.

Regarding existing software that the reviewer mentioned – CaImAn (Giovannucci et al. 2019): We evaluated CaImAn, which is a powerful framework designed for large-scale, multicellular calcium imaging (e.g., motion correction, denoising, and automated cell/ROI extraction). However, it is not optimized for the per-event kinetics central to our project - such as extracting rise and decay times for individual quantal events at single synapses. Achieving this level of granularity would typically require additional custom Python scripting and parameter tuning within CaImAn's code-centric interface. This runs counter to CaFire's design goals of a nocode, task-focused workflow that enables users to analyze miniature events quickly and consistently without specialized programming expertise.

Regarding Igor Pro (WaveMetrics), (Müller et al. 2012): Igor Pro is another platform that can be used to analyze calcium imaging signals. However, it is commercial (paid) software and generally requires substantial custom scripting to fit the specific analyses we need. In practice, it does not offer a simple, open-source, point-and-click path to per-event kinetic quantification, which is what CaFire is designed to provide.

The authors should be commended for making their software publicly available, but there are some questions:

How does CaFire compare to existing tools?

As mentioned above, we have not been able to adapt the custom scripts used by various labs for our purposes, including software developed in MatLab (Blum et al. 2021), Python (Xing and Wu 2018), and Igor (Müller et al. 2012). Some in the field do use semi-publicly available software, including Nikon Elements (Chen and Huang 2017) and CaImAn (Giovannucci et al. 2019). However, these platforms are not optimized for the per-event kinetics central to our project - such as extracting rise and decay times for individual quantal events at single synapses. We have added more details about CaFire, mainly focusing on the workflow and measurements, highlighting the superiority of CaFire, showing that CaFire provides a no-code, standardized pipeline with automated miniature-event detection and per-event metrics (e.g., amplitude, rise time τ , decay time τ), optional $\Delta R/R$ support, and auto-partition feature. Collectively, these features make CaFire simpler to operate without programming expertise, more transparent and reproducible across users, and better aligned with the event-level kinetics required for this project.

Very few details about the Huygens deconvolution algorithms and input settings were provided in the methods or text (outside of MLE algorithm used in STED images, which was not Ca²⁺ imaging). Was it blind deconvolution? Did the team distill the point-spread function for the fluorophores? Were both channels processed for ratiometric imaging? Were the same settings used for each channel? Importantly, please include SVI Huygens in the 'Software and Algorithms' Section of the methods.

We thank the reviewer for raising this important point. We have now expanded the Methods to describe our use of Huygens in more detail and have added SVI Huygens Professional (Scientific Volume Imaging, Hilversum, The Netherlands) to the “Software and Algorithms” section. For Ca²⁺ imaging data, time-lapse stacks were processed in the Huygens Deconvolution Wizard using the standard estimation algorithm (CMLE). This is not a blind deconvolution procedure. Instead, Huygens computes a theoretical point-spread function (PSF) from the full acquisition metadata (objective NA, refractive index, voxel size/sampling, pinhole, excitation/emission wavelengths, etc.); if refractive index values are provided and there is a mismatch, the PSF is adjusted to account for spherical aberration. We did not experimentally distill PSFs from bead measurements, as Huygens’ theoretical PSFs are sufficient for our data.

Both green (GCaMP) and red (mScarlet) channels were processed for ratiometric imaging using the same workflow (stabilization, optional bleaching correction, and deconvolution within Huygens). For each channel, the PSF, background, and SNR were estimated automatically by the same built-in algorithms, so the underlying procedures were identical even though the numerical values differ between channels because of their distinct wavelengths and noise characteristics. Importantly, Huygens normalizes each PSF to unit total intensity, such that the deconvolution itself does not add or remove signal and therefore preserves intensity ratios between channels; only background subtraction and bleaching correction can change absolute fluorescence values. For the mScarlet channel, where we observed modest bleaching (~1.10 over 15 sec), we applied Huygens’ bleaching correction and visually verified that similar structures maintained comparable intensities after correction. For presynaptic GCaMP signals, bleaching over these short recordings was negligible, so we omitted the bleaching-correction step to avoid introducing multiplicative artifacts. This workflow ensures that ratiometric $\Delta R/R$ measurements are based on consistently processed, intensity-conserving deconvolved images in both channels.

The number of deconvolution iterations could have had an effect when comparing GCaMP series; please provide an average number of iterations used for at least one experiment. For example, Figure 3, Syt::GCaMP6s, Scar8f & Scar8m, and, if applicable, the maximum number of permissible iterations.

We thank the reviewer for this comment. For all Ca²⁺ imaging datasets, deconvolution in Huygens was performed using the recommended default settings of the CMLE algorithm with a maximum of 30 iterations. The stopping criterion was left at the Huygens default, so the algorithm either converged earlier or, if convergence was not reached, terminated at this 30-iteration limit. No other iteration settings were used across the GCaMP series (lines 555-559).

Please clarify if the 'Express' settings in Huygens changed algorithms or shifted input parameters.

We appreciate the reviewer’s question regarding the Huygens “Express” settings. For clarity, we note that all Ca²⁺ imaging data reported in this manuscript were deconvolved using the “Deconvolution Wizard”, not the “Deconvolution Express” mode. In the Wizard, we explicitly selected the CMLE algorithm (or GMLE in a few STED-related cases as recommended by SVI), using the recommended maximum of 30 iterations, and other recommended settings while allowing Huygens to auto-estimate background and SNR for each channel. Bleaching

correction was toggled manually per channel (applied to mScarlet when bleaching was evident, omitted for GCaMP when bleaching was negligible), as described in the revised Methods (lines 553-559).

By contrast, the Deconvolution Express tool in Huygens is a fully automated front-end that can internally adjust both the choice of deconvolution algorithm (e.g., CMLE vs. GMLE/QMLE) and key input parameters such as SNR, number of iterations, and quality threshold based on the selected “smart profile” and the image metadata. In preliminary tests on our datasets, Express sometimes produced results that were either overly smoothed or showed subtle artifacts, so we did not use it for any data included in this study. Instead, we relied exclusively on the Wizard with explicitly controlled settings to ensure consistency and transparency across all GCaMP series and ratiometric analyses.

We suggest including a sample data set, perhaps in Excel, so that future users can beta test on and organize their data in a similar fashion.

We agree that this would be useful, a point shared by R1 above. In response, we have added a sample data set to the GitHub site and included sample ImageJ data along with screenshots to explain the analysis in more detail. These improvements are discussed in the manuscript (lines 705-708).

(3) While the challenges of AZ imaging are mentioned, it is not discussed how the authors tackled each one. What is defined as an active zone? Active zones are usually identified under electron microscopy. Arguably, the limitation of GCaMP-based sensors targeted to individual AZs, being unable to resolve local Ca²⁺ changes at individual boutons reliably, might be incorrect. This could be a limitation of the optical setup being used here. Please discuss further. What sensor performance do we need to achieve this performance level, and/or what optical setup would we need to resolve such signals?

We appreciate the reviewer’s thoughtful comments and agree that the technical challenges of active zone (AZ) Ca²⁺ imaging merit further clarification. We defined AZs, as is the convention in our field, as individual BRP puncta at NMJs. These BRP puncta co-localize with individual puncta of other AZ components, including CAC, RBP, Unc13, etc. ROIs were drawn tightly over individual BRP puncta and only clearly separable spots were included.

To tackle the specific obstacles of AZ imaging (small signal volume, high AZ density, and limited photon budget at high frame rates), we implemented both improved sensors and optimized analysis (Fig. 6). First, we introduced a ratiometric AZ-targeted indicator, BRP::mScarlet3::GCaMP8m (Bar8m), and computed $\Delta R/R$ with $\Delta R/R$ with $R(t) = F_{GCaMP8m}/F_{mScarlet3}$. ROIs were drawn over individual AZs (Fig. 6B). Under our standard resonant area-scan conditions (~118 fps), Bar8m produces robust $\Delta R/R$ transients at individual AZs (example peaks ≈ 3.28 ; $\tau_{rise} \approx 9.0$ ms; Fig. 6C, middle), indicating that single-AZ signals can be detected reproducibly when AZs are optically resolvable.

Second, we increased temporal resolution using high-speed Galvano line-scan imaging (~1058 fps), which markedly sharpened the apparent kinetics ($\tau_{rise} \approx 3.23$ ms) and revealed greater between-AZ variability (Fig. 6C, right; 6D–E). Population analyses show that line scans yield much faster rise times than area scans (Fig. 6D) and a dramatically higher fraction of significantly different AZ pairs (8.28% and 4.14% in 8f and 8m areascans vs 78.62% in 8m line-scan, lines 721-725), uncovering pronounced AZ-to-AZ heterogeneity in Ca²⁺ signals. Together, these revisions demonstrate that under our current confocal configuration, AZ-targeted GCaMP8m can indeed resolve local Ca²⁺ changes at individual, optically isolated boutons.

We have revised the Discussion to clarify that our original statement about the limitations of AZ-targeted GCaMPs refers specifically to this combination of sensor and optical setup, rather than an absolute limitation of AZ-level Ca²⁺ imaging. In our view, further improvements in baseline brightness and dynamic range ($\Delta F/F$ or $\Delta R/R$ per action potential), combined with

sub-millisecond kinetics and minimal buffering, together with optical configurations that provide smaller effective PSFs and higher photon collection (e.g., higher-NA objectives, optimized 2-photon or fast line-scan modalities, and potentially super-resolution approaches applied to AZ-localized indicators), are likely to be required to achieve routine, high-fidelity Ca^{2+} measurements at every individual AZ within a neuromuscular junction.

(4) In Figure 5: Only GCaMP8f (Bar8f fusion protein) is tested here. Consider including testing with GCaMP8m. This is particularly relevant given that GCaMP8m was a more successful GECI for subcellular post-synaptic imaging in Figure 6.

We appreciate this point and request by Reviewer 3. The main limitation for detecting local calcium changes at AZs is the speed of the calcium sensor, and hence we used the fastest available (GCaMP8f) to test the Bar8f sensor. While replacing GCaMP8f with GCaMP8m would indeed be predicted to enhance sensitivity (SNR), since GCaMP8m does not have faster kinetics relative to GCaMP8f, it is unlikely to be a more successful GECI for visualizing local calcium differences at AZs.

That being said, we agree that the Bar8m tool, including the improved mScarlet3 indicator, would likely be of interest and use to the field. Fortunately, we had engineered the Bar8m sensor while this manuscript was in review, and just recently received transgenic flies. We have evaluated this sensor, as requested by the reviewer, and included our findings in Fig. 1 and 6. In short, while the sensitivity is indeed enhanced in Bar8m compared to Bar8f, the kinetics remain insufficient to capture local AZ signals. These findings are discussed in the revised manuscript (lines 424-442, 719-730), and we appreciate the reviewer for raising these important points.

In earlier experiments, Bar8f yielded relatively weak fluorescence, so we traded frame rate for image quality during resonant area scans (~60 fps). After switching to Bar8m, the signal was bright enough to restore our standard 118 fps area-scan setting. Nevertheless, even with dual-channel resonant area scans and ratiometric (GCaMP/mScarlet) analysis, AZ-to-AZ heterogeneity remained difficult to resolve. Because Ca^{2+} influx at individual active zones evolves on sub-millisecond timescales, we adopted a high-speed singlechannel Galvano line-scan (~1 kHz) to capture these rapid transients. We first acquired a brief area image to localize AZ puncta, then positioned the line-scan ROI through the center of the selected AZ. This configuration provided the temporal resolution needed to uncover heterogeneity that was under-sampled in area-scan data. Consistent with this, Bar8m line-scan data showed markedly higher AZ heterogeneity (significant AZ-pair rate ~79%, vs. ~8% for Bar8f area scans and ~4% for Bar8m area scans), highlighting Bar8m's suitability for quantifying AZ diversity. We have updated the text, Methods, and figure legend accordingly (tell reviewer where to find everything).

(5) Figure 5D and associated datasets: Why was Interquartile Range (IQR) testing used instead of ZScoring? Generally, IQR is used when the data is heavily skewed or is not normally distributed. Normality was tested using the D'Agostino & Pearson omnibus normality test and found that normality was not violated. Please explain your reasoning for the approach in statistical testing. Correlation coefficients in Figures 5 E & F should also be reported on the graph, not just the table. In Supplementary Table 1. The sub-table between 4D-F and 5E-F, which describes the IQR, should be labeled as such and contain identifiers in the rows describing which quartile is described. The table description should be below. We would recommend a brief table description for each sub-table.

Thank you for this helpful suggestion. We have updated the analysis in two complementary ways. First, we now perform paired two-tailed t-tests between every two AZs within the same preparation (pairwise AZ-AZ comparisons of peak responses). At $\alpha < 0.05$, the fraction of significant AZ pairs is ~79% for Bar8m line-scan data versus ~8% for Bar8f area-scan data,

indicating markedly greater AZ-to-AZ diversity when measured at high temporal resolution. Second, for visually marking the outlying AZs, we re-computed the IQR (Q1–Q3) based on the individual values collected from each AZs (15 data points per AZ, 30 AZs for each genotype), and marked AZs whose mean response falls above Q3 or below Q1; IQR is used here solely as a robust dispersion reference rather than for hypothesis testing. Both analyses support the same observation: Bar8m line-scan data reveal substantially higher AZ heterogeneity than Bar8f and Bar8m area-scan data. We have revised the Methods, figure panels, and legends accordingly (t-test details; explicit “IQR (Q1–Q3)” labeling; significant AZ-pair rates reported on the plots) (lines 719–730).

(6) Figure 6 and associated data. The authors mention: 'SynapGCaMP quantal signals appeared to qualitatively reflect the same events measured with electrophysiological recordings (Fig. 6D).' If that was the case, shouldn't the ephys and optical signal show some sort of correlation? The data presented in Figure 6D show no such correlation. Where do these signals come from? It is important to show the ROIs on a reference image.

We apologize this was not clear, as similar points were raised by R1 and R2. We were just showing separate (uncorrelated) sample traces of electrophysiological and calcium imaging data. Given how confusing this presentation turned out to be, and the fact that we show the correlated ephys and calcium imaging events in Fig. 7, we have elected to remove the uncorrelated electrophysiological events in Fig. 6 to just focus on the calcium imaging events (now Figures 7 and 8).

Figure 7B: Were Ca²⁺ transients not associated with mEPSPs ever detected? What is the rate of such events?

This is an astute question. Yes indeed, during simultaneous calcium imaging and current clamp electrophysiology recordings, we occasionally observed GCaMP transients without a detectable mEPSP in the electrophysiological trace. This may reflect the detection limit of electrophysiology for very small minis; with our noise level and the technical limitation of the recording rig, events < ~0.2 mV cannot be reliably detected, whereas the optical signal from the same quantal event might still be detected. The fraction of calcium-only events was ~1–10% of all optical miniature events, depending on genotype (higher in lines with smaller average minis). These calcium-only detections were low-amplitude and clustered near the optical threshold (lines 361–365).

Minor comments

(1) It should be mentioned in the text or figure legend whether images in Figure 1 were deconvolved, particularly since image pre-processing is only discussed in Figure 2 and after.

We thank the reviewer for pointing this out. Yes, the confocal images shown in Figure 1 were also deconvolved in Huygens using the CMLE-based workflow described in the revised Methods. We applied deconvolution to improve contrast, reduce out-of-focus blur, and better resolve the morphology of presynaptic boutons, active zones, and postsynaptic structures, so that the localization of each sensor is more clearly visualized. We have now explicitly stated in the Fig. 1 legend and Methods (lines 575–577) that these images were deconvolved prior to display.

(2) The abbreviation, SNR, signal-to-noise ratio, is not defined in the text.

We have corrected this error and thank the reviewer for pointing this out.

(3) Please comment on the availability of fly stocks and molecular constructs.

We have clarified that all fly stocks and molecular constructs will be shared upon request (lines 747-750). We are also in the process of depositing the new Scar8f/m, Bar8f/m, and SynapGCaMP sensors to the Bloomington Drosophila Stock Center for public dissemination.

(4) Please add detection wavelengths and filter cube information for live imaging experiments for both confocal and widefield.

We thank the reviewer for this helpful suggestion. We have now added the detection wavelengths and filter cube configurations for both confocal and widefield live imaging to the Methods.

For confocal imaging, GCaMP signals were acquired on a Nikon A1R system using the FITC/GFP channel (488-nm laser excitation; emission collected with a 525/50-nm band-pass filter), and mScarlet signals were acquired using the TRITC/mCherry channel (561-nm laser excitation; emission collected with a 595/50-nm band-pass filter). Both channels were detected with GaAsP detectors under the same pinhole and scan settings described above (lines 512-517).

For widefield imaging, GCaMP was recorded using a GFP filter cube (LED excitation ~470/40 nm; emission ~525/50 nm), which is now explicitly described in the revised Methods section (lines 632-633).

(5) Please include a mini frequency analysis in Supplemental Figure S1.

We apologize for not including this information in the original submission. This is now included in the Supplemental Figure S1.

(6) In Figure S1B, consider flipping the order of EPSP (currently middle) and mEPSP (currently left), to easily guide the reader through the quantification of Figure S1A (EPSPs, top traces & mEPSPs, bottom traces).

We agree these modifications would improve readability and clarity. We have now re-ordered the electrophysiological quantifications in Fig. S1B as requested by the reviewer.

(7) Figure 6C: Consider labeling with sensor name instead of GFP.

We agree here as well, and have removed “GFP” and instead added the GCaMP variant to the heatmap in Fig. 7C.

(8) Figure 6E, 7B, 7E: Main statistical differences highlighting sensor performance should be represented on the figures for clarity.

We did not show these differences in the original submission in an effort to keep the figures “clean” and for clarity, putting the detailed statistical significance in Table S1. However, we agree with the reviewer that it would be easier to see these in the Fig. 6E and 7B,E graphs. This information has now been added the Figs. 7 and 8.

(9) Please report if the significance tested between the ephys mini (WT vs IIB^{-/-}, WT vs IIA^{-/-}, IIB^{-/-} vs IIA^{-/-}) is the same as for Ca²⁺ mini (WT vs IIB^{-/-}, WT vs IIA^{-/-}, IIB^{-/-} vs IIA^{-/-}). These should also exhibit a very high correlation (mEPSP (mV) vs Ca²⁺ mini deltaF/F). These tests would significantly strengthen the final statement of “SynapGCaMP8m can capture physiologically relevant differences in quantal events with similar sensitivity as electrophysiology.”

We agree that adding the more detailed statistical analysis requested by the reviewer would strengthen the evidence for the resolution of quantal calcium imaging using SynapGCaMP8m. We have included the statistical significance between the ephys and calcium minis in Fig. 8

and included the following in the revised methods (lines 358-361), the Fig. 8 legend and Table S1:

Using two-sample Kolmogorov–Smirnov (K–S) tests, we found that SynapGCaMP8m Ca^{2+} minis ($\Delta F/F$, Fig. 8E) differ significantly across all genotype pairs (WT vs IIB^{-/-}, WT vs IIA^{-/-}, IIB^{-/-} vs IIA^{-/-}; all $p < 0.0001$). The genotype rank order of the group means ($\pm$ SEM) is IIB^{-/-} > WT > IIA^{-/-} (0.967 ± 0.036 ; 0.713 ± 0.021 ; 0.427 ± 0.017 ; $n=69, 65, 59$). For electrophysiological minis (mEPSP amplitude, Fig. 8F), K–S tests likewise show significant differences for the same comparisons (all $p < 0.0001$) with D statistics of 0.1854, 0.3647, and 0.4043 (WT vs IIB^{-/-}, WT vs IIA^{-/-}, IIB^{-/-} vs IIA^{-/-}, respectively). Group means ($\pm$ SEM) again follow IIB^{-/-} > WT > IIA^{-/-} (0.824 ± 0.017 mV; 0.636 ± 0.015 mV; 0.383 ± 0.007 mV; $n=41$ each). These K–S results demonstrate identical significance and rank order across modalities, supporting our conclusion that SynapGCaMP8m resolves physiologically relevant quantal differences with sensitivity comparable to electrophysiology.

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